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一、一种基于数据驱动和可解释人工智能的电信行业客户流失预测方法

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译文：

摘要

在竞争激烈的电信行业（TCI）中，客户留存对于盈利至关重要，因为客户流失本身仍然还是一个重大挑战。传统机器学习（ML）模型往往缺乏对复杂电信数据所需的预测能力，而黑箱模型则提供有限的透明度，降低信任度和可操作的见解。本研究引入了 XAI-Churn TriBoost，这是一个可解释的数据驱动模型，使用超过 200 万条记录的数据集开发而成。该模型通过软投票集成结合极端梯度提升（XGBoost）、类别梯度提升（CatBoost）和轻量级梯度提升机（LightGBM），以增强客户流失预测。数据预处理包括通过贝叶斯岭回归进行迭代插补来处理缺失值。序列数据缩放通过结合鲁棒缩放、标准缩放和最小-最大缩放方法来实现，以确保特征的一致性。特征选择使用随机森林（RF）进行的 Boruta 技术，训练数据中的类别不平衡问题则通过合成少数类过采样技术（SMOTE）加以解决。XAI-Churn TriBoost 实现了高预测性能，准确率为 96.44%，精确率为 92.82%，召回率为 87.82%，F1 得分为 90.25%。为了增强模型的透明度，我们引入了可解释的人工智能（AI）技术，具体包括局部可解释模型无关解释（LIME）和夏普利加法解释（SHAP），以解释个体预测并识别影响客户流失的关键特征。影响客户流失的关键因素包括规律性和金额，为 TCI 提供了有价值的见解，以制定有针对性的客户保留策略。因此，XAI-Churn TriBoost 不仅提供了强大的性能，还具备可解释性，突显了其在支持 TCI 客户保留工作中的潜力。

关键词

客户流失预测，电信行业，留存策略，决策制定，商业智能，数据科学，机器学习，预测分析，可解释人工智能

1. 引言

现代通信高度依赖 TCI，它推动着创新、全球商业和交流。随着市场趋于饱和，竞争日益激烈，客户留存的重要性已不亚于客户获取。价格战、消费者期望的变化以及频繁的技术变革是企业在当今高度竞争环境中面临的诸多挑战之一。对于电信运营商而言，客户留存已成为战略重点，尤其是在运营成本上升和监管约束日益严格的背景下，保持高质量服务变得愈发困难。竞争性、运营商数量、

服务种类等指标都促成了 TCI 在过去几十年中的显著增长和发展。然而，激烈的竞争、饱和的市场以及不断变化的环境也带来了重大挑战，尤其是在管理客户流失方面[60]。饱和的市场环境使客户能够轻松地在服务和提供商之间转换，而高度竞争的服务种类进一步加剧了这一问题。 从服务提供商的角度来看，流失的客户要么停止使用其服务，要么转向竞争对手，这会对公司的财务表现产生重大影响 [56]。

客户流失是指客户停止使用服务，这会直接影响盈利能力，降低品牌忠诚度，增加客户获取成本，并侵蚀收入。高流失率还会干扰资源分配和网络规划，影响服务的连续性。研究表明，获取新客户可能比维持现有客户花费五到二十五倍的成本，因此降低流失率在经济上至关重要 [10]。全球化和技术进步进一步给 TCI 带来压力，因为它们为客户提供更多选择 [15]。日益激烈的竞争、价格监管、市场整合以及 OTT 服务的流行，使收入和盈利能力受到更严重的冲击 [45]，[51]。在这一背景下，识别关键客户行为、提升客户参与度以及保持竞争优势都依赖于精确的客户流失预测。

客户流失是 TCI 非常关注的问题，因为留住现有客户比寻找新客户要便宜得多[52]。长期留住消费者对于保持收入的稳定流动和在竞争中保持领先地位至关重要。客户离开企业的最常见原因是长时间内对所获得的服务不满意。很多时候，人们会在逐渐对服务感到不满后决定停止使用。服务提供商可以通过主动测量客户满意度并识别服务改进的领域来应对这一问题。通过区分忠诚客户和可能流失的客户，电信公司可以运用预测算法来制定针对性的挽留策略。预测客户流失对于 TCI 来说是一个重要的重点，因为它直接影响品牌价值和收入产出。电信企业通常会详细记录客户数据和服务使用情况，以便识别模式并优化挽留策略。准确预测客户流失对于维持 TCI 的增长至关重要，因为客户保留具有极其重要的意义[32]。

在各个领域，机器学习已被证明具有良好的适应性[25]，[2]。机器学习已成为电信行业（TCI）中预测消费者行为和在大型数据集中识别模式的强大工具，从而降低客户流失率。客户风险评估和流失预测都广泛使用了集成技术和其他复杂的机器学习模型。这些模型使电信公司能够开展有针对性的客户挽留措施，高效分配资源，并优化其营销策略。通过将运营方式从被动应对转变为主动预防，机器学习改变了企业与客户互动的方式，并有助于客户留存[29]，[22]。尽管传统机器学习模型有其潜力，但它们的预测结果往往不够准确且难以理解[40]。尽管这些模型能够生成准确的预测，但利益相关者却难以信任复杂的集成模型和其他不透明的算法[50]。在像电信行业（TCI）这样的行业中，实际发现是战略决策的基础，缺乏透明度可能会阻碍机器学习解决方案的采用[42]。 为弥补这一缺陷，可解释人工智能提供了一种理解模型输出、了解各属性重要性并生成准确预测的

方法。使用在准确性和可解释性之间取得平衡的可解释人工智能，可以做出更优且更透明的客户流失管理决策 [8]，[61]。

为应对电信行业客户流失问题，本研究遵循关键概念。我们预计，一个强大的预处理流程将提供一致且高质量的数据，这将为精确建模奠定基础。XAI-Churn TriBoost 模型在预测性能方面预计会优于现有最先进的模型。该模型结合了 XGBoost、LightGBM 和 CatBoost，采用投票集成。我们希望通过使用 LIME 和 SHAP 等可解释人工智能方法，提高清晰度和理解力，揭示客户流失的原因并提供解决方案。该模型在包含超过 200 万条记录的大数据集上进行训练和测试，其中 60%用于训练，40%用于测试，旨在确保模型的稳健性和可扩展性。通过结合先进的建模方法和可解释性，本研究意在提供一个数据驱动的解决方案，支持 TCI 识别高风险客户并实施针对性的客户留存策略，从而确保高准确性和可操作的洞察力，以支持战略决策。

论文其余部分的结构如下。第 2 节回顾了相关工作，指出了以往研究中的不足，并解释了解决这些问题的重要性。第 3 节描述了方法论，包括数据处理、机器学习技术以及可解释人工智能的使用。第 4 节展示了实验结果，评估了模型性能并讨论了关键发现。第 5 节对工作进行了详细讨论，比较了不同模型，分析了以往研究，并提出了未来研究的方向。最后，第 6 节总结了研究成果并提出了未来工作的建议。

2. 相关研究

已有各种研究探讨了电信行业客户流失预测，采用了多样化的机器学习技术和方法。Ullah 等人[54]提出了一个用于数据分析的客户流失模型，并通过常见的评估指标进行了验证。他们提出的模型表现出更好的性能。随机森林（RF）达到了 88%的准确率。作者还进行了聚类分析，以识别影响流失的主要变量。他们最后为电信决策者提供了可行的建议，以保留现有客户。

Ammar 等人 [1] 引入了一种混合算法方法，以有效预测流失客户。尽管所建议的混合萤火虫算法与传统萤火虫算法的准确性相似，但混合算法在处理时间上的极短延迟时表现要好得多。作者使用了 F1 分数、准确率、精确率和召回率等指标来评估这些算法。他们强调，电信行业中的预测模型必须在精确率和效率之间取得平衡 [19]。创建更小、更具维度的训练模型可能是提高性能的一种直接方法。特征提取和选择的方法尤其有效，能够在减少计算负担的情况下实现准确的预测。数据挖掘在多个研究中已显示出其在 TCI 中预测客户流失的价值 [28]。很多研究都探讨了诸如 k 近邻（KNNs）、决策树（DTs）和随机森林（RF）等分类器的有效性 [46]。

许多使用机器学习方法预测客户流失的模型已经被开发出来[21]，[12]。利用这些算法，TCI 可以识别客户即将流失的时间，并提供更好的服务以吸引他们

继续留下。基于决策树和随机森林的策略实施提高了电信服务的盈利能力。为了处理缺失数据，预测均值匹配方法被采用，而不仅仅是删除特征或忽略缺失值的数据[57]。为了应对大规模数据集、特征标签转换和数据分布不均的问题，TCI 已在客户流失预测模型中引入集成分类器。根据 Saran Kumar 等人[36]对客户流失预测方法的全面分析，许多现有模型未能准确预测未来的客户流失。为了保持客户忠诚度，他们强调了精确预测模型的重要性。

一种改进的数据挖掘应用构建技术由[34]提出，为他们后续的发展奠定了基础。他们的方法通过应用数据挖掘技术来预测预付费用户留存率得到了验证。这些研究突显了预测建模、特征工程和性能评估在有效进行电信行业客户流失分析中的重要性。[27]提供了客户有意识地分析导致客户流失和利润特征的案例。结果表明，改进后的模型显著优于其原始的非优化版本。在评估的模型中，使用多项式核的支持向量机（SVM）表现最佳，取得了超过 84%的 F1 分数和超过 97%的准确率。Alzubaidi 等人[4]利用通话详单记录特征来衡量已识别社区内用户之间的社交关系强度。他们通过在通话图上应用客户流失传播模型，评估了流失节点的净聚合流失影响。 逻辑回归（LR）被用来纳入流失影响效应，并预测特定用户的流失趋势。本研究强调，通过利用社交关系属性，如通话和短信详情，分析用户连接可以增强流失预测模型。

Khalid 等人 [33] 比较了各种预测模型，发现决策树（DTs）的表现优于其他方法。他们的研究结果证实，决策树是客户流失预测中的顶级方法之一。同样地，[58] 旨在利用逻辑回归（LR）和大数据分析预测 TCI 中的高价值客户流失。通过分析历史客户数据，该研究在客户库中识别出可能流失的客户，并基于流失行为特征支持有针对性的挽回措施。Lalwani 等人 [38] 比较了几种用于客户流失预测的机器学习方法，包括逻辑回归（LR）、朴素贝叶斯（NB）、支持向量机（SVM）、决策树（DTs）、随机森林（RF）、XGBoost、CatBoost、自适应提升（adaptive boosting）以及极端树分类器（extra tree classifier）。他们的实验结果表明，集成算法，如自适应提升和 XGBoost，是最准确的模型，实现了 84%的曲线下面积（AUC），并在准确率、精确率、召回率和 F1 分数等指标上表现出色。

在回顾了这些研究之后，一些局限性变得显而易见。许多研究依赖于小数据集，这限制了其发现的普遍性和稳健性。所使用的机器学习技术往往过时或缺乏复杂性，对先进的集成方法或混合方法关注极少。数据预处理也是一个关键的不足之处，因为大多数研究在处理缺失值时仅依赖简单的填补方法，而没有采用先进的特征工程或选择技术来提升模型性能。此外，类别不平衡问题很常见，这通常导致预测结果有偏，而测试数据集过小进一步降低了这些模型在应用于新数据时的可靠性。可解释人工智能方法的引入也严重不足，这对于理解和解释模型预

测至关重要。电信行业中的战略决策高度依赖于可解释人工智能所提供的可操作见解。由于这些局限性， 流失预测的研究需要更强大、更具扩展性且更易解释的模型。为了解决这些不足，采取了以下创新性步骤：

1. 构建了一个稳健的预处理流程，采用贝叶斯岭回归的迭代填补方法有效处理缺失值。

2. 该方法依次结合鲁棒缩放、标准缩放和最小-最大缩放来对数据进行缩放，同时保持其分布完整性。

3. 为提高模型的准确性和可解释性，使用基于随机森林的 Boruta 特征选择方法来识别重要特征。

4. 建议使用一个结合 XGBoost、LightGBM 和 CatBoost 的投票分类器，以提高 XAI-Churn TriBoost 模型的预测准确性和可靠性。

5. 使用 60%的数据进行训练，40%用于测试，模型在一个包含超过 200 万条记录的大数据集上进行了训练和评估，证明了其在大数据集上的有效性和鲁棒性。

6. 为了解释模型的预测结果并提供对影响客户流失变量的实用理解，采用了 LIME 和 SHAP 等可解释性方法。

3. 方法论

如图 1 所示，我们提出的 XAI-Churn TriBoost 模型用于预测客户流失，遵循一个系统化的流程。首先收集流失数据集。然后通过贝叶斯岭回归进行迭代填补的预处理步骤对数据进行清洗、编码和缩放。接着，我们使用基于随机森林的 Boruta 技术进行特征工程，生成新特征并选择重要特征。数据被划分为训练集和测试集，并使用 SMOTE 对训练集进行平衡。我们使用三种 Boosting 算法（XGBoost、CatBoost 和 LightGBM）的软投票集成方法来训练模型。为了实现可解释性，我们应用了可解释人工智能技术，包括 LIME 和 SHAP，以理解预测结果并提供可操作的见解。

3.1. 数据集

在本研究中，我们使用了通过 Zindi 平台上的 Expresso Churn Prediction Challenge 提供的数据集[59]。本研究使用的是 train.csv 数据集，该数据集对参加 Zindi 平台竞赛的参与者开放。该数据集包含数值型和类别型特征，共有 17 个变量和 2,154,048 个样本。这些变量涵盖了客户行为、购买以及在网络中的互动等各类细节。目标变量“churn”表示客户是否在连续 90 天内停止使用服务且无任何交易记录。我们使用 60%的数据来训练模型，40%的数据用于测试我们提出的模型。

3.2. 数据预处理

数据清洗、编码、缺失值填补和缩放都是本节所述数据集预处理的一部分。

3.2.1. 数据清洗

数据清洗是必要的，因为数据集中包含大量缺失值（如图 2 所示）以及一些无用的特征。我们删除了以下特征：zone1 和 zone2，它们表示对电信网络内特定区域的通话，由于它们的缺失值超过 90%，因此不适合分析。moving to rivaval group (MRG) 特征，表示客户即将离开，也被删除，因为它只有一个唯一值。user_id 是一个匿名标识符，被删除因为它对预测分析没有贡献。region 特征，用于标识客户的位置，被排除以使模型适用于不同国家，并避免引入基于位置的偏差。top_pack，表示客户使用最流行的套餐，以及 freq_top_pack，记录最流行套餐的激活次数，由于其高基数—包含 140 个分类值—并且套餐类型在不同电信公司之间差异很大，限制了其普遍适用性，因此也被删除。删除这些特征后，我们保留了 11 个在电信公司中普遍存在的特征，以便进一步分析。

3.2.2. 数据编码

数据编码是机器学习中的关键步骤，因为模型通常使用数值型数据。分类特征由非数值型值组成，需要转换为模型可以解释的格式。为此，我们使用序数编码器来处理我们的分类特征。序数编码器将分类值转换为整数标签。由于每个类别都被分配了一个特定的数字，模型可以轻松地处理每个特征中的唯一类别。当类别具有内在顺序时，这种策略尤为有效。为了使模型能够从分类数据中正确学习，序数编码器将值转换为数值形式。

3.2.3. 数据填补

由于缺失值会显著影响分析的质量，需要采用适当的方法来处理缺失数据。当数据缺失量较大时，仅仅删除包含缺失信息的记录可能相当不切实际。在某些情况下，删除数据可能导致重要信息的大量丢失 [20]。当缺失数据较少时，传统的填补方法如使用均值、中位数或众数填补空缺是有效的 [18]。然而，在本例中我们遇到了大量的缺失数据，因为某些特征的缺失值超过了 50%。为了解决这个问题，采用了迭代填补方法。

在统计学中，迭代插补是一种通过反复将具有缺失数据的特征建模为其他特征的函数来估计缺失值的技术。通过反复改进插补值并利用特征间的相互依赖关系，该方法提高了精度[39]。我们在整个重复插补过程中主要使用贝叶斯岭模型作为估计器。该线性回归模型的系数包含先验分布，从而实现了正则化。结果更加一致，过拟合也得到了更好的控制。通过估计不确定性特征，插补方法能够稳健地应对数据噪声和不可预测性，并且易于实现。

由于我们的数据集包含异常值，我们通过使用中位数值来填补缺失值来开始迭代填补过程。这有助于减少异常值的影响，因为中位数比均值对异常值的敏感度更低。在这一步之后，迭代填补继续进行最多二十次迭代。我们通过设置绝对容差为 0.00001 来引导收敛。该方法会结合这一容差水平以及数据的尺度来确

定一个缩放后的容差。在我们的案例中，算法采用了一个大约为 18.24 的缩放容差作为收敛阈值。经过 15 次迭代填补后，填补的值稳定在缩放容忍度水平内，表明已经收敛。因此，缺失值的填补过程是彻底且可靠的。在算法 1 中，展示了处理缺失数据的算法。

3.2.4. 数据缩放

数据缩放是机器学习中的关键步骤，因为它可以提高模型的准确性并提升学习效率。我们的数据集中包含大量异常值，因此需要一种谨慎的缩放方法来有效处理这些异常值，同时不扭曲数据分布。

为实现这一目标，我们采用了一种序列数据缩放方法[6]。首先，我们使用鲁棒缩放来减少异常值的影响。与删除这些异常值不同，我们基于中位数和四分位距（IQR）对它们进行缩放。这使得缩放对极端值的敏感度降低。鲁棒缩放的公式为：

$$Robust\ Scaling = \frac{X - median(X)}{IQR(X)} \tag{1}$$

在此方法中，值围绕中位数进行中心化，相比基于均值的方法，可以减少异常值的影响。

在进行鲁棒缩放后，我们应用标准缩放来标准化特征。这种缩放方法将数据调整为均值为 0，标准差为 1。标准缩放使所有特征处于同一尺度，从而提升模型性能。标准缩放的公式如下：

$$Standard\ Scaling = \frac{X - \mu}{\sigma} \tag{2}$$

此处，μ表示特征的均值，σ是其标准差。这种方法对于对数据大小敏感的模型非常有用，因为它将每个特征对齐到一个标准化的范围内。

最后，我们应用最小-最大缩放来将特征归一化到一个固定的范围内，通常在 0 到 1 之间。这确保了在模型训练过程中所有特征都能平等贡献，尤其是在特征范围对算法有重要影响的情况下。最小-最大缩放的公式如下：

$$MinMax\ Scaling = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{3}$$

其中X_{min}和X_{max}分别是特征 X 的最小值和最大值。这种归一化过程在保留数据内部关系的同时，将其压缩到标准范围内。

我们成功处理了异常值，并通过我们的顺序缩放策略使数据集准备好以实现最佳模型性能。

3.3. 特征工程

机器学习依赖于特征工程。引入新特征或修改已有特征对于提升模型至关重要。有效的特征工程可以揭示数据集中的隐藏关系。这不仅提高了模型的预测能力，也使其更容易理解。为了使模型更好地学习模式，我们从输入数据中提取有用的特征并加以使用。

我们的数据库包含两个不同的特征：tigo 和 orange。orange 特征显示拨打 orange 网络的通话，而 tigo 特征显示拨打 tigo 网络的通话。这两个特征记录相同的数据，反映了拨打不同网络的出站通话。为了简化数据集，我们将它们合并为一个特征，代表拨打外部网络的总通话量。我们定义这个新特征为：

$$\mathrm{Off_{net}} = \mathrm{tigo} + \mathrm{orange} \tag{4}$$

通过将通话详单与这个新的离网特征结合起来，我们有可能简化数据集。因此，模型能够更好地识别显著的模式。通过将所有离网互动纳入模型，而不是一次只考虑一个网络，使模型更加全面。

3.4. 特征选择

特征选择是机器学习中的关键部分，它通过从数据集中提取有用的属性来提高模型的性能和可解释性。通过提升模型准确性、简化模型以及同时降低维度，可以防止过拟合。特征选择使模型更容易理解和解释，通过专注于相关变量来简化模型。具体参考[16]。

特征选择可以使用多种方法；然而，这些方法通常依赖于阈值或领域知识进行手动选择，这可能导致偏差。虽然更传统的做法有助于识别特征，但它们有可能忽略关键的交互作用或依赖主观判断。我们采用 Boruta 特征选择方法来解决这个问题，该方法通过将原始特征与随机生成的“影子”特征进行比较，是一种自动化且有效的策略。Boruta 通过反复评估特征的重要性，从而增强选择过程的客观性和可靠性，仅保留真正相关的特征[37]。表 1 展示了在 Boruta 特征选择策略中使用的超参数及其取值。

在每次迭代中，Boruta 算法依赖于诸如准确率均值减少量等指标来确定特征的相关性。它使用基于树的分类器作为其基础模型。我们使用 RF 分类器构建 Boruta，因为其集成学习方法能够很好地评估特征的相对价值。为了提高稳定性并减少方差，模型设置了 100 个估计器，这意味着它由 100 棵决策树组成。90 百分位数确保只有重要性得分高于至少 90%的影子特征的特征才会被保留。我们将显著性水平设为 0.05，建立了一个严格的阈值以过滤掉不相关的特征。为了在性能和效率之间取得平衡，最大迭代次数被设为 100，以防止不必要的计算。此外，将随机状态设为 42 确保了结果的一致性和可重复性。这种方法系统地剔除不相关的特征，只保留最有意义的特征。通过结合 Boruta 与 RF 模型，我们选择了一个可靠的特征子集，从而提高了模型的整体性能和可解释性。

3.5. 数据分割

数据分割对于正确评估模型和实现泛化至关重要。以下方程表示将数据集划分为两个部分：

$$D = D_{training} + D_{testing} \tag{5}$$

在本研究中，我们将数据划分为 60%用于训练（$D_{training}$）和 40%用于测试

（$D_{testing}$）。我们选择这个比例主要是为了降低过拟合的风险。当模型在数据集的大部分上进行训练时，它可能会过于适应训练集，从而无法很好地泛化到新数据。通过将 40%的数据分配给测试集，我们可以更稳健地评估模型的性能。由于我们的方法结合了多个 Boosting 模型，更大的测试集有助于更有效地评估模型的泛化能力。Boosting 模型虽然强大，但如果在没有适当验证的情况下使用过多数据进行训练，就容易出现过拟合。此外，我们的数据集高度不平衡，仅有 19%的样本代表流失情况。使用更大的测试集可以确保我们有足够的流失样本来准确衡量模型在少数类上的性能。

为保持一致性并确保可重复性，我们将随机状态设置为 42。我们还应用了分层抽样，以保持训练集和测试集中的原始类别分布。图 3 展示了原始数据集、训练子集和测试子集中的流失分布。在表 4 中，我们提供了训练数据的统计描述，在表 5 中，我们提供了测试数据的统计描述。这些表格提供了关于数据分布、变化范围和规模的见解。

3.6. 数据平衡

在机器学习应用中，不平衡数据集很常见。当数据集不平衡时，少数类只占整体数据的一小部分。这种不平衡会使模型预测偏向多数类，从而影响整体性能。为了解决这个问题，已经开发了几种重采样技术[30]。传统方法包括复制少数类的样本或减少多数类样本的数量以平衡数据集。然而，这些方法可能会引入诸如过拟合或丢失有价值信息等问题，因此在许多情况下效果不佳。重要的是，重采样应仅应用于训练数据集。如果应用于整个数据集，可能会导致模型产生偏差，因为它会改变真实世界的分布。

SMOTE 方法[13] 是处理类别不平衡的一个流行策略。通过为少数类创建合成样本，SMOTE 避免了基本过采样所带来的过拟合风险，后者涉及复制现有的样本。该方法通过创建一个平衡或几乎平衡的训练数据集，从而提升模型在两个类别上的表现[11]。SMOTE 通过将少数类中的两个相近样本x和x_R组合来创建合成样本。在少数类中，从x的邻居中随机选择最近邻x_R。合成样本的计算方式如下：

$$x_{synthetic} = x + u \times (x_R - x) \tag{6}$$

其中$0 \leq u \leq 1$是一个随机值。

为了解决我们数据集中的类别不平衡问题，我们使用了 imbalanced-learn Python 库中的 SMOTE 方法。SMOTE 仅在训练数据上进行实现，并使用随机状态 42 以确保可重复性。该算法使用了 5 个最近邻，意味着合成样本是基于 5 个最接近的少数类样本生成的。采样策略设置为 0.5，使少数类样本数量增加到多数类样本数量的 50%。最初，少数类在训练数据中占 19%，在应用 SMOTE 后，这一比例增加到了 33%，如图 4 所示。表 6 展示了使用 SMOTE 平衡后

的训练数据的统计摘要。它说明了在生成合成样本后，数据的分布、变异性和规模发生了怎样的变化。

3.7. 我们提出的 XAI-Churn TriBoost

我们提出的模型 XAI-Churn TriBoost 结合了三种 Boosting 算法：XGBoost、CatBoost 和 LightGBM。这些算法被集成到一个统一的软投票集成框架中，用于预测客户流失。Boosting 算法在处理复杂模式方面非常有效。使用集成方法可以利用每个模型的优势，从而提升整体性能。以下是对我们方法的逐步解释，如 Algorithm 2 所示。

XAI-Churn TriBoost 集成模型由三个基础模型组成。第一个模型 XGBoost 是一种流行的梯度提升算法，以其可扩展性和通过先进的正则化技术对过拟合的鲁棒处理而闻名[14]。它能够处理大型数据集和非线性关系[44]。XGBoost 使用决策树作为弱学习器，并通过 L1 和 L2 正则化对其进行优化。这种正则化有助于防止过拟合，使模型在未见过的数据上具有更好的泛化能力。XGBoost 的目标函数定义为：

$$Obj = \sum_{i=1}^{m} l\left(y_i, \hat{y}_i^{(t-1)} + f_i(x_i)\right) + \Omega(f_k) \tag{7}$$

这里，l表示损失函数，y_i是真实标签，$\hat{y}_i^{(t-1)}$是第$(t-1)$次迭代的预测结果。样本x_i的函数输出表示为$f_i(x_i)$。正则化项$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\omega^2$用于控制模型的复杂度。

第二个基模型 CatBoost 是专门设计用来高效处理分类数据，而无需进行大量预处理[48]。通过其递归训练技术，CatBoost 在每棵后续树中减少误差，从而提高准确性和稳定性，同时降低预测偏差[24]。

此外，LightGBM 由于其快速的训练速度和低内存消耗，非常适合处理大数据集。与传统的基于层级的树生长方法不同，LightGBM 优先生长信息量最大的叶子，并逐个叶子地生长树。使用这种基于 LightGBM 的方法可以提高准确性，加快收敛速度，同时减少所需的树的数量。为了降低内存使用并提高处理性能，LightGBM 采用基于直方图的方法将连续变量离散化为区间。特别是在处理高维数据时，这种方法进一步提升了性能 [31]。

我们采用随机搜索交叉验证（randomized search，CV）进行超参数调优，以确保所有模型都在最佳性能状态下运行。我们的广泛数据集增强了策略的有效性。通过随机选择超参数，它能够高效地探索庞大的参数空间，而无需分析所有可能的组合，如[9]所述。我们为每个模型的超参数指定一个可能的取值范围，并允许其尝试一定数量的组合。这种方法在探索与效率之间找到了平衡点，非常适合复杂且高维的参数空间。每个模型最终选定的超参数见表 7。

我们使用一种软投票技术来整合每个模型训练后的预测结果。通过加权概率，每个类别的模型输出可以通过软投票来确定。在我们的 XAI-Churn TriBoost 模型中，类别 c 的最终预测概率用 $P_{TriBoost}(c)$ 表示。该概率是通过使用特定权重将基础模型的预测概率进行组合计算得出的。计算方式如下：

$$P_{TriBoost}(c) = \frac{\omega_1 \cdot P_{XGBoost}(c) + \omega_2 \cdot P_{CatBoost}(c) + \omega_3 \cdot P_{LightGBM}(c)}{\omega_1 + \omega_2 + \omega_3} \tag{8}$$

在这个公式中，$P_{XGBoost}(c)$、$P_{CatBoost}(c)$ 和 $P_{LightGBM}(c)$ 分别代表 XGBoost、CatBoost 和 LightGBM 模型预测的类别 c 的概率。权重 ω_1、ω_2和 ω_3表示每个模型在集成中的相对重要性。我们分别为 XGBoost、CatBoost 和 LightGBM 分配权重 $\omega_1 = 2$、$\omega_2 = 1$ 和 $\omega_3 = 3$。这些权重反映了每个模型的性能特征。

每个样本的最终预测类别是根据 $P_{TriBoost}(c)$ 中的最高概率选择的。这种方法确保选择的是模型间达成最强共识的类别。这种方法提高了预测的可靠性，并增强了模型的整体鲁棒性。

3.8. 可解释的人工智能

简化复杂 AI 模型并揭示其预测逻辑的方法属于可解释人工智能领域[26]。这在理解模型决策能提升信心和实际洞察力的领域中尤为重要，例如客户流失预测。我们通过使用 LIME 和 SHAP 来实现这一点。

3.8.1. 局部可解释的模型无关解释

复杂 ML 模型可以通过 LIME 更好地理解。基于单个预测，它构建一个更简单且易于理解的模型。其核心概念是针对模型进行局部近似，仅关注特定实例周围的区域。这有助于阐明初始模型所做出预测的推理过程。LIME 在实例周围生成扰动样本，然后使用这些样本训练一个可解释的模型（如线性模型），以反映原始模型在该实例附近的行为 [49]。

LIME 通过最小化一个损失函数来工作，该函数衡量原始模型 h 与可解释模型 q 在目标实例 x 周围的差异。近似过程通过接近度 $\pi(x')$进行加权。目标函数如下：

$$\delta(x) = \arg\min_{q \in Q}\big(M(h, q, \pi) + \Lambda(\mathrm{q})\big) \tag{9}$$

这里，$M(h, q, \pi)$量化了简单模型 q 在局部上对 h 的近似程度。$\Lambda(q)$项用于惩罚复杂性，以保持可解释性。LIME 会为更接近目标实例的样本赋予更高的权重，从而确保局部一致的解释。这种方法可以为文本、图像和表格数据提供可操作的见解 [7]。

3.8.2. 夏普利加法解释

SHAP 是一种流行的可解释人工智能技术。它提供了对模型预测中每个特征贡献的洞察。根据合作博弈论的原则，SHAP 为每个属性分配一个夏普利值。该值代表了该特征对预测的贡献。对于复杂的机器学习模型，SHAP 真正表现出色，因为透明度至关重要。它将模型视为一个谜，并解释单个预测以及特征的总体价值，参考 [41]。

特征 j 的夏普利值$\phi(j)$是通过在所有可能的特征组合中平均其边际贡献来计算的。对于特征集合 F 和模型 v，夏普利值$\phi(j)$由以下公式给出：

$$\phi(j) = \frac{1}{|F|!} \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \big(v(S\ U\{j\}) - v(S)\big) \tag{10}$$

在这个公式中，S 表示 F 中排除 j 的每个子集。$v(S\ U\{j\}) - v(S)$项计算将 j 添加到子集 S 中所产生的影响。SHAP 通过累加所有组合的贡献来提供每个特征影响的统计量。这种方法在局部和全局范围内都提供了可解释性 [3]。

4. 结果

4.1. 环境和设置

4.1.1. 编程语言和库

本研究选择了 Python 作为主要语言，因为其适应性强、易于使用且拥有丰富的库支持。在数据的分析和处理方面，使用了 Pandas，而 NumPy 则促进了高效数值运算和数组结构。使用 Scikit-learn 构建机器学习模型、进行数据预处理和缺失值填补，并评估模型性能。Seaborn 被用来创建数据可视化图表，确保其既美观又具有信息量。在特征选择方面，采用了 Boruta 库来识别并保留最相关的特征。为了解决数据不平衡问题，使用了 imblearn 库，特别是通过 SMOTE 方法实现类别分布的平衡。

各种 Boosting 库，包括 LightGBM、CatBoost 和 XGBoost，通过先进的梯度提升技术实现了稳健的模型开发。使用 Scikit-learn 创建了一个投票分类器，以结合多个模型的预测结果。为了实现模型可解释性，采用了 LIME 和 SHAP 等可解释性库，提供了对特征重要性的局部和全局见解，从而加深了对模型预测的理解。

4.1.2. 硬件规格

所有实验和模型训练均在 HP ZBook 15 G6 移动工作站上进行。该笔记本电脑配备了 Intel Core i9-9880H 处理器、NVIDIA Quadro T2000 显卡、32GB DDR4 内存以及 512GB SSD。

4.2. 性能评估指标

本研究中使用的性能评估指标包括准确率、精确率、召回率、F1 分数以及曲线分析指标，如接收者操作特征曲线下的面积（AUC-ROC）和精确率-召回率

（PR）曲线。此外，这些指标的计算还参考了混淆矩阵中的术语，包括真正例（TP）、真反例（TN）、假正例（FP）和假反例（FN）。在客户流失预测任务中，TP 表示正确预测的流失实例，TN 表示正确预测的非流失实例，FP 表示错误预测为流失的非流失实例，FN 表示错误预测为非流失的流失实例。图 5 展示了混淆矩阵，呈现了 XAI-Churn TriBoost 模型的预测结果。

4.2.1. 准确率

准确率是正确预测实例数与总实例数的比率。它反映了 XAI-Churn TriBoost 模型的整体预测准确性。准确率通过公式计算：

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

图 6 比较了我们的模型与其他基础模型的准确率。XAI-Churn TriBoost 表现出更高的准确率，表明其性能更优。

4.2.2. 精确率

精确度是指正确识别的流失实例占所有预测为流失实例的比例。它衡量了模型避免误报的能力。精确度由公式计算：

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

图 7 展示了我们的模型和基准模型的精确度得分。XAI-Churn TriBoost 模型表现出更高的精确度，反映了其在减少误判流失情况方面的有效性。

4.2.3. 召回率

召回率是正确识别的流失案例与所有实际流失案例的比例。它衡量模型检测所有流失实例的能力。召回率的计算如公式所示：

$$Recall = \frac{TP}{TP + FN} \tag{13}$$

图 8 展示了 XAI-Churn TriBoost 模型与基础模型的召回率得分。XAI-Churn TriBoost 模型表现出更高的召回率，表明其在捕捉大多数流失案例方面更为有效。

4.2.4. F1 分数

F1 分数是精确率和召回率的调和平均值。它平衡了模型进行精确且全面预测的能力。F1 分数通过公式计算：

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

图 9 展示了基础模型和 XAI-Churn TriBoost 模型的 F1 分数。XAI-Churn TriBoost 模型取得了较高的 F1 分数，这表明它有效地处理了误报和漏报。

4.2.5. 曲线分析

我们利用曲线分析中的两个关键指标 AUC-ROC 和 PR 曲线来考察 XAI-Churn TriBoost 模型的性能。借助这些指标，可以更好地理解该模型在面对不同

分类标准集合时的性能表现。

模型在所有阈值下的性能度量是 AUC-ROC。ROC 曲线显示，分类阈值的逐一变化会对真正率和假正率产生影响。模型在阈值为 1 时表现最佳；在 0.5 时，其预测结果基本上是随机的。PR 曲线对于流失预测和其他不平衡数据集非常有益，它展示了在不同分类阈值下准确率和召回率之间的相关性。该图表明，即使类别不平衡，模型仍能保持良好的召回率和精确率。

XAI-Churn TriBoost 模型的 AUC-ROC 分数为 0.9828，PR AUC 分数为 0.9601，如图 10 所示。该模型的近乎完美的结果表明，它能够有效区分流失和非流失事件，并且在召回率和精确率之间的平衡得到了良好维护。这验证了该模型能够以较少的误报成功识别流失案例，并具有良好的分类性能。

4.3. 可解释性分析与可解释人工智能技术

可解释人工智能在理解复杂模型方面变得至关重要。在客户流失预测等应用中，了解模型为何做出特定预测非常重要。这种透明度有助于建立对模型决策的信任和信心。传统的机器学习模型虽然可解释，但往往缺乏处理复杂电信数据所需的预测能力。为了解决这一问题，采用先进的可解释人工智能方法，如 LIME 和 SHAP。这些技术可以解释单个预测并突出特征的重要性。它们使流失预测模型更具可解释性，为利益相关者提供清晰且可操作的见解。

4.3.1. LIME 分析

图 11 展示了我们模型做出的流失预测的 LIME 解释。右侧显示了特定实例的特征值。中间部分展示了模型的决策边界。用蓝色突出的特征有助于“不流失”预测，而用橙色突出的特征则有助于“流失”预测。左侧显示了模型的预测结果，表明流失概率为 53%，实际类别也为流失。影响此预测的关键特征包括收入、规律性和 ARPU 分段。收入和规律性对流失有较强的贡献，推动预测结果朝向流失。相比之下，ARPU 分段和金额则略微支持“不流失”预测。然而，橙色特征的整体影响导致最终预测为流失。

图 12 展示了一个 LIME 解释，其中模型预测客户不会流失的概率为 78%，这与实际的无流失类别一致。在此情况下，收入对流失有贡献，但像规律性、ARPU 分段和数据量等特征的综合影响更倾向于无流失。这些蓝色贡献加强了无流失的预测。

4.4. SHAP 分析

图 13 展示了一个力图，提供了该实例中每个特征对流失预测的初步概览。该模型预测流失概率为 0.53。在这个图中，增加流失可能性的特征显示为红色，而减少流失可能性的特征显示为蓝色。对于该实例，关键特征如‘regularity’和‘revenue’（红色）推动了流失的预测，而‘apru_segment’（蓝色）则降低了流失的可能性。

为了获得每个特征影响的更详细分解，使用了图 14 中的瀑布图。该图显示了每个特征对最终预测分数的确切数值影响。在这里，‘regularity’（+0.21）和‘revenue’（+0.06）是对流失预测最强的贡献因素，而‘apru_segment’（-0.03）则会降低该预测。

这些图表共同提供了模型为何预测该特定实例会发生客户流失的全面视图，其中力图（force plot）提供总体概览，而瀑布图（waterfall plot）则详细展示了特征对预测结果的贡献。

从更广泛的角度来看，图 15 中的 SHAP beeswarm 图展示了每个特征的 SHAP 值分布，说明其在多个实例中的影响。每个点代表一个实例，其在 x 轴上的位置表示该实例的 SHAP 值。从蓝色到红色的颜色渐变反映了特征值，其中红色表示较高的值，蓝色表示较低的值。例如，‘regularity’（红色）的高值与客户流失可能性的强烈正向影响相关，推动预测结果向流失方向偏移。该图揭示了特征影响的模式和方向，其中像‘regularity’和‘montant’这样的特征具有最大的影响。

图 16 中的 SHAP 特征重要性图根据平均绝对 SHAP 值对特征进行排序，展示了它们对模型预测的整体贡献。‘Regularity’和‘montant’是影响客户流失预测的前两大特征，其次是‘arpu segment’和‘revenue’。此图提供了客户流失关键驱动因素的清晰概览，有助于模型解释的关键因素识别。

这些 SHAP 总结图表整体来看，揭示了对模型预测具有持续影响的特征及其影响方向，有助于理解特征在局部和全局范围内的价值。

4.5. 消融研究

在这里，我们展示了使用 XAI-Churn TriBoost 模型进行消融研究的结果，用于评估 XGBoost、CatBoost 和 LightGBM 的各种权重组合的影响。在 XAI-Churn TriBoost 模型中，通过合并这三种 Boosting 算法创建了一个单一的软投票集成模型。软投票利用每个类别模型的加权概率输出来确定最终预测。为了确保最佳性能，评估每种模型的贡献并调整其权重是至关重要的。我们通过将 XGBoost、CatBoost 和 LightGBM 的权重从 1 到 3 进行变化，开展了实验并记录了每种组合的性能指标。表 8 总结了此次消融研究的结果。在分析了所有权重组合的性能后，我们选择了 XGBoost: 2，CatBoost: 1，LightGBM: 3 的配置，因为该组合在多个评估标准下均表现出更优的性能。

5. 讨论

客户流失是电信行业（TCI）中的一个重要问题，因为留住客户对盈利至关重要。机器学习有潜力在大型数据集中识别模式，使企业能够提前预测客户流失。这使企业能够在客户离开之前采取战略性措施来保留客户。机器学习在这些行业中已被证明能显著提高企业的盈利能力。然而，其中一个关键挑战是机器学习模

型需要大量数据，并且通常作为“黑箱”运作，使得行业专业人士难以完全理解其预测结果。这种缺乏可解释性的现象会引发信任问题，因此可解释的人工智能变得尤为重要。

在本研究中，我们提出了一种名为 XAI-Churn TriBoost 的模型，该模型通过软投票集成 XGBoost、CatBoost 和 LightGBM。本研究基于一个包含超过两百万条记录的数据集进行，其中 60%的数据用于训练，40%用于测试。该数据集中包含大量缺失值，我们采用基于贝叶斯岭回归的迭代插补方法来处理，而不是传统的插补技术。此外，该数据集还存在大量异常值和特征范围差异。为了解决这一问题，我们采用顺序缩放方法，首先使用鲁棒缩放来减少异常值的影响，然后进行标准化，最后使用最小-最大缩放将数据归一化到 0 到 1 的范围内。

最初，数据集包含 17 个特征，但在经过全面的数据清洗、去除无关特征和特征工程处理后，我们将特征数量减少到 10 个。为了确保仅保留最相关和重要的特征，我们采用 Boruta 特征选择方法，这是一种自动化且有效的技术，通过将原始特征与随机生成的“影子”特征进行比较来实现。该过程剔除了一个特征，最终留下 9 个选定的特征用于建模。在建模过程中，我们使用基于树的算法，包括 XGBoost、CatBoost 和 LightGBM，因为它们非常适合处理大型且复杂的数据集。为了优化模型性能，我们应用随机搜索交叉验证来进行超参数调优。随后，这三个模型的预测结果通过软投票集成方法进行结合，从而提高整体的准确性和鲁棒性。

我们的模型优于各个基础模型，达到了 0.9644 的准确率，0.9282 的精确率，0.8782 的召回率，以及 0.9025 的 F1 分数。此外，我们将我们的方法与几种最先进的模型进行了比较，包括决策树（DTs）、随机森林（RF）、逻辑回归（LR）、高斯朴素贝叶斯（Gaussian NB）、伯努利朴素贝叶斯（Bernoulli NB）、岭分类器（Ridge Classifier）和随机梯度下降（Stochastic Gradient Descent）。为了确保公平比较，所有这些模型都应用了相同的预处理步骤。结果如表 9 所示，高斯朴素贝叶斯由于其概率分类方法和特征独立性的假设，实现了最高的召回率（0.9672）。这一特性使其更有可能将边缘案例分类为正类，从而减少假阴性并提高召回率。然而，这种改进是以较低的精确率和整体准确率为代价的。

相比之下，我们提出的 XAI-Churn TriBoost 模型在整体表现上最佳，实现了最高的准确率（0.9644）、精确率（0.9282）和 F1 分数（0.9025）。虽然诸如决策树（DT）、岭分类器和随机梯度下降（Stochastic Gradient Descent）等模型在某些方面表现良好，但它们在所有评估指标上并不保持同样的平衡。XAI-Churn TriBoost 模型的优势在于其能够有效识别流失客户同时减少误报，使其成为专注于客户保留策略的企业最可靠和实用的选择。

由于可解释性在这一领域至关重要，我们引入了 LIME 和 SHAP。这些技

术在个体预测层面和整体模型层面都提供了可解释性，帮助利益相关者理解预测背后的推理。SHAP 分析表明，规律性和金额是影响客户流失的最重要特征。规律性指的是过去三个月内的活动次数，而金额代表充值金额。基于这些发现，可以看出流失风险高度依赖于消费者的使用行为模式和交易量。

在本研究中，我们的重点仅限于 TCI 的客户流失预测建模。然而，近年来，许多研究已经探讨了不同行业中客户流失的预测，包括银行和金融服务、电子商务和零售，以及流媒体和娱乐行业。这些研究应用了不同的机器学习技术，以识别影响客户流失的因素并改进客户保留策略。

一个例子是[5]为泰国的一家软件即服务（SaaS）库存管理公司创建的流失预测模型。在实施了多种机器学习模型后，他们发现当使用 10 折交叉验证时，随机森林（RFs）表现最佳，召回率为 0.916，F1 分数为 0.926。同样地，[17]在葡萄牙的一家软件公司构建了一个预测模型，以识别可能离职的员工。根据他们的研究结果，XGBoost 在 ROC AUC 为 0.86 和召回率为 0.85 的情况下优于其他方法。

在印度班加罗尔，[23] 相关研究团队创建了一个模型来预测在线食品配送行业的客户流失。他们发现，在所有评估的机器学习模型中，朴素贝叶斯（NB）和随机森林（RFs）分类器的 AUC 得分为 0.952，表现最佳。根据[53]，在银行业开发了一个基于 XGBoost 的流失预测模型，结果准确率为 0.839，灵敏度为 0.601，特异度为 0.903，F1 分数为 0.613。同样地，[55] 使用 LightGBM 应用了多个机器学习模型，取得了准确率为 0.91，精确率为 0.94，召回率为 0.87，F1 分数为 0.90 的成果。

游戏行业也对客户流失预测进行了研究。[43] 创建了一个用于预测免费移动视频游戏玩家流失的模型。他们的结果表明，RFs 模型在预测流失时间方面准确率范围为 0.66 到 0.95，预测的时间范围为 1 至 7 天。

电信行业 TCI 领域的客户流失预测一直是多项研究的主题，每项研究都使用了独特的数据集和机器学习方法。例如，[47]使用了包含 20 个特征的 7,043 个样本，采用随机梯度提升方法，取得了 0.981 的准确率，0.928 的精确率和 0.927 的召回率。类似地，[35]在包含 15 个特征的 2,000 个样本上使用 RFs 模型，也达到了 0.981 的准确率，0.928 的精确率和 0.927 的召回率。

讨论的研究突显了流失预测模型在多个行业中的多样化应用，每个行业都采用不同的机器学习技术与数据集。为了提供一个简洁的比较，表 10 总结了这些研究，概述了各自所属的行业、方法和主要发现。

在这项研究中，我们的主要关注点是 TCI 中的客户流失预测。我们利用一个包含超过 200 万条记录的单一大规模数据集来开发和评估我们提出的模型。然而，我们计划在后续工作中扩展分析，通过在多个行业中评估模型的有效性，

并整合更多的数据集来进一步验证。这将确保我们的方法不仅适用于 TCI 行业，还能帮助确定其相对优势和不足。

我们的目标是使用其他地区的数据来评估模型，从而考虑客户行为的区域差异，并进一步提高其泛化能力。此外，我们还打算构建一个网页应用，可以连接到实时数据源并具备监控功能。企业将能够通过此应用程序实时监控重要指标，评估客户行为，并测量客户流失率。这将使他们能够做出主动决策。

应用程序的比较分析功能将使企业能够根据人口统计学特征、产品使用情况和行为趋势对客户进行分类。这一工具将帮助企业识别最可能取消订阅的消费者群体，从而制定有针对性的客户保留策略。我们计划实施一个反馈循环，使用户能够提供对模型预测的见解，以确保模型的持续改进。该模型将定期利用积累的反馈进行再训练，以适应客户行为的变化并提高其预测准确性。

该模型可以通过不断添加新数据并识别趋势和模式保持相关性，因此可以应对外部因素，如经济动荡或市场变化，影响消费者偏好的情况。此外，自动化这一过程将通过提高预测准确性并减少对人工干预的依赖来增强适应性。最终，这种方法提高了模型的准确性，从而有助于决策制定、客户留存策略的定制以及行业的长期可靠性。

6. 结论

本研究的目的是提出一种用于电信行业客户流失预测的模型，称为 XAI-Churn TriBoost。该集成模型利用软投票机制，结合了 XGBoost、CatBoost 和 LightGBM。通过使用超过 200 万条记录的数据集，确保了数据质量。该模型在 158 万条记录上进行训练，在超过 86 万条记录上进行测试，取得了 F1 分数为 0.9025、准确率为 0.9644、精确率为 0.9282、召回率为 0.8782 的优异表现。整合可解释人工智能方法提高了模型的透明度。使用 LIME 和 SHAP 来解释单个预测并突出特征重要性，从而提高了可解释性并建立了利益相关者的信任。SHAP 分析表明，规律性和金额是影响客户流失的最重要特征，为留存策略提供了可操作的见解。XAI-Churn TriBoost 的出色表现突显了在解决客户流失预测挑战中可解释模型的重要性。

在后续工作中，我们计划通过引入不同行业的多个数据集来评估模型的泛化能力。此外，我们还计划开发一个基于网络的应用程序，实现实时监控，以跟踪流失率和客户行为，从而促进主动决策。还将实施一个反馈循环，通过定期使用新的用户洞察数据重新训练模型来优化预测，确保模型能够适应不断变化的趋势并保持持续的预测准确性。

附：原文：

Abstract

In the competitive telecommunications industry (TCI), retaining clients is crucial

for profitability, as customer churn remains a significant challenge. Traditional machine learning (ML) models often lack the predictive power needed for complex telecom data, while black-box models provide limited transparency, reducing trust and actionable insights. This study introduces XAI-Churn TriBoost, an interpretable and explainable data-driven model developed using a dataset of over 2 million records. The model combines extreme gradient boosting (XGBoost), categorical boosting (CatBoost), and light gradient boosting machine (LightGBM) in a soft voting ensemble to enhance churn prediction. Data preprocessing included handling missing values through iterative imputation with a Bayesian ridge. Sequential data scaling was implemented by combining robust, standard, and min-max scaling methods to ensure feature consistency. Feature selection was conducted using the Boruta technique with a random forest (RF), and class imbalance in the training data was addressed using the synthetic minority oversampling technique (SMOTE). XAI-Churn TriBoost achieved high predictive performance, with an accuracy of 96.44%, precision of 92.82%, recall of 87.82%, and F1 score of 90.25%. To enhance model transparency, we incorporated explainable artificial intelligence (AI) techniques, specifically local interpretable model-agnostic explanations (LIME) and Shapley additive explanations (SHAP), to interpret individual predictions and identify critical features affecting churn. Key factors impacting churn include regularity and montant, offering TCI valuable insights for targeted retention strategies. XAI-Churn TriBoost thus provides both robust performance and interpretability, highlighting its potential to support customer retention efforts in the TCI.

Keywords

Churn prediction；Telecom industry；Retention strategies；Decision-making；Business intelligence；Data science；Machine learning；Predictive analytics；Explainable artificial intelligence

1. Introduction

Modern connectivity relies heavily on the TCI, which drives innovation, global business, and communication. As markets become saturated and competition intensifies, customer retention has become as crucial as customer acquisition. Price wars, evolving consumer expectations, and frequent technological disruptions are among the challenges businesses face in today's highly competitive environment. For telecommunication carriers, customer retention is a strategic priority, especially as maintaining high-quality service becomes increasingly difficult amidst rising operational costs and growing regulatory constraints. Competitiveness, the number of

operators, service offerings, and other metrics have all contributed to the TCI remarkable growth and development over the past several decades. However, intense competition, saturated markets, and a constantly evolving environment have also led to significant challenges, particularly in managing customer churn [60]. The saturated market landscape allows customers to easily switch between services and providers, and the wide variety of highly competitive service offerings further exacerbates the issue. From the perspective of service providers, churned customers either discontinue using their services or switch to competitors, resulting in a substantial impact on the company's financial performance [56].

Customer churn, where customers stop using a service, directly impacts profitability by reducing brand loyalty, increasing customer acquisition costs, and eroding revenue. High churn rates also disrupt resource allocation and network planning, compromising service continuity. Research shows that it might cost five to twenty-five times more to acquire a new customer than to maintain an existing one, hence reducing churn is economically vital [10]. Globalization and technical improvements put further strain on TCI since they give customers more alternatives [15]. Revenue and profitability have been hit even harder by rising competition, pricing regulation, market consolidation, and the popularity of over-the-top services [45], [51]. Recognizing critical customer behaviors, improving customer engagement, and maintaining competitive advantage all depend on precise churn prediction in this setting.

Customer churn is a big concern for TCI because keeping current customers is much cheaper than finding new ones [52]. Keeping consumers around for the long run is crucial for maintaining a constant flow of revenue and remaining ahead of the competition. The most common reason for clients to abandon a firm is when they are unhappy with the services they receive for a long time. Most of the time, people decide to cease using a service after becoming increasingly unhappy with it over time. Service providers can address this by proactively measuring client satisfaction and identifying areas for service improvement. By separating loyal customers from those at risk, telecom companies can employ predictive algorithms to develop tailored retention tactics. Predicting customer turnover is an important emphasis for TCI as it has a direct impact on brand value and revenue output. It is common practice for telecoms to meticulously document client data and service usage in order to identify patterns and refine retention strategies. Maintaining TCI's growth depends on accurately predicting customer churn, since client retention is of utmost importance [32].

In various domains, ML has proven its adaptability [25], [2]. ML has emerged as

a powerful tool in TCI for predicting consumer behavior and identifying patterns in large datasets, therefore lowering customer attrition. Customer risk assessment and churn prediction have both seen extensive use of ensemble techniques and other sophisticated ML models. The models allow TCI to conduct targeted retention initiatives, allocate resources efficiently, and fine-tune their marketing strategies. By shifting from reactive to proactive operations, ML alters the manner in which firms engage with customers and aids in client retention [29], [22]. Even though conventional ML models have promise, their predictions are often inaccurate and difficult to understand [40]. Stakeholders struggle to trust complex ensemble models and other opaque algorithms, despite the fact that they generate accurate forecasts [50]. In industries like TCI, where practical findings form the basis of strategic decisions, the absence of transparency can hinder the adoption of ML solutions [42]. To compensate for this shortcoming, explainable AI offers a way to understand model output, understand the importance of various attributes, and generate accurate predictions. Use explainable AI, which balances accuracy with interpretability, to make customer churn management decisions that are better and more transparent [8], [61].

In order to combat client turnover in the telecom sector, this study adheres to essential concepts. We anticipate that a strong preprocessing pipeline will provide consistent and high-quality data, which will set the stage for precise modeling. The XAI-Churn TriBoost model is expected to outperform state of the art models in terms of predictive performance. It combines XGBoost, LightGBM, and CatBoost in a voting ensemble. We hope to improve clarity and understanding by using explainable AI methods like LIME and SHAP to shed light on the causes of customer attrition and offer solutions. Trained and tested on a large dataset of over 2 million records—with 60% allocated for training and 40% for testing—the model is designed to ensure robustness and scalability. Through the combination of advanced modeling and explainability, this study seeks to deliver a data-driven solution that supports TCI in identifying at-risk customers and implementing targeted retention strategies, ensuring both high accuracy and actionable insights for strategic decision-making.

The rest of the paper is structured as follows. Section 2 reviews related work, highlights gaps in previous research and explains the importance of addressing these issues. Section 3 describes the methodology, including data processing, ML techniques, and the use of explainable AI. Section 4 presents experimental results, evaluates model performance and discusses key findings. Section 5 provides a detailed discussion of the work, compares different models, analyzes previous studies, and outlines directions for

future research. Finally, Section 6 concludes with a summary of contributions and suggestions for future work.

2. Related work

Various studies have explored customer churn prediction in the TCI, employing diverse ML techniques and methodologies. Ullah et al. [54] presented a customer attrition model for data analytics, which was validated using common assessment measures. Their proposed model demonstrated improved performance. RF achieved an 88% accuracy. The authors also performed cluster profiling to identify the major variables influencing churn based on churn risk. They concluded by offering actionable advice for telecom decision-makers on retaining their existing customers.

Ammar et al. [1] introduced a hybridized algorithm approach to effectively forecast churners. Although the accuracy of the suggested hybrid firefly algorithm and the traditional firefly algorithm were similar, the hybrid algorithm performed far better when dealing with very brief delays in time. F1 score, accuracy, precision, and recall were some of the metrics used to assess the algorithms by the writers. They emphasized that forecast models for the telecom industry must strike a compromise between precision and efficiency [19]. Creating smaller, more dimensional training models can be a straightforward way to improve performance. Methods for feature extraction and selection were especially effective, enabling accurate predictions with less computing load. Data mining has demonstrated its worth in churn prediction within the TCI in several studies [28]. A lot of research has looked into the effectiveness of classifiers such as k-nearest neighbors (KNNs), decision trees (DTs), and RF [46].

Many models that use ML approaches to forecast client churn have been developed [21], [12]. Using these algorithms, the TCI can figure out when customers are about to depart and give them better service to entice them to stay. The implementation of DT- and RF-based strategies has increased the profitability of telecom services. Instead of simply removing features or disregarding data with missing values, predictive mean matching has been used to handle missing data [57]. To deal with huge datasets, shifting feature labels, and uneven data distributions, the TCI has included ensemble classifiers into consumer speculation models. According to Saran Kumar et al. [36], who conducted a comprehensive analysis of methods for predicting customer attrition, many existing models fail to accurately anticipate future churn. In order to maintain client loyalty, they emphasized the importance of precise prediction models.

An improved technique for constructing data mining applications was proposed by [34], providing the foundation for their subsequent development. Their approach

was validated through the application of data mining technologies to predict prepaid subscriber retention. These studies underscore the importance of predictive modeling, feature engineering, and performance evaluation for effective customer churn analysis in the TCI. Examples of customers who consciously decided to analyze the traits and behaviors that lead to customer churn and profits were provided by [27]. The results demonstrated that enhanced models significantly outperformed their plain, non-optimized counterparts. Among the models evaluated, support vector machine (SVM) with polynomial kernel emerged as the best performer, achieving an F1 score of over 84% and an accuracy of more than 97%. Alzubaidi et al. [4] employed call detail record features to measure the strength of user-to-user social relationships within identified communities. Using a churn propagation model on the call graph, they assessed the net aggregated churn influence from churner nodes. Logistic regression (LR) was utilized to incorporate the churn influence effect and predict the churn tendency of specific users. This study highlighted that examining user connections enhances churn prediction models by leveraging social relationship attributes, such as call and SMS details.

Khalid et al. [33] compared various prediction models, and found that DTs outperformed other methods. Their findings confirmed that DTs are among the top methods for churn prediction. Similarly, [58] aimed to predict high-value customer turnover in the TCI using LR and big data analytics. By analyzing historical customer data, the study identified probable churners in the customer library and supported targeted win-back efforts based on churn behavior characteristics. Lalwani et al. [38] compared several ML methods for churn prediction, including LR, naïve bayes (NB), SVM, DTs, RF, XGBoost, CatBoost, adaptive boosting, and extra tree classifier. Their experimental results showed that ensemble algorithms, such as adaptive boosting and XGBoost, were the most accurate models, achieving an area under the curve of 84% and excelling across metrics like accuracy, precision, recall, and F1 score.

After reviewing these studies, several limitations become evident. Many rely on small datasets, limiting the generalizability and robustness of their findings. The ML techniques used are often outdated or lack sophistication, with minimal focus on advanced ensemble methods or hybrid approaches. Data preprocessing is another critical shortfall, as most studies rely on simple imputation methods for handling missing values instead of employing advanced feature engineering or selection techniques to improve model performance. Furthermore, class imbalance problems are common, which frequently results in biassed predictions, and testing datasets that are too small further reduce these models' reliability when applied to new data. The

incorporation of explainable AI approaches, which are crucial for comprehending and making sense of model predictions, is also severely lacking. Strategic decision-making in the telecom sector relies heavily on actionable insights provided by explainable AI. Research into churn prediction needs better models that are more resilient, scalable, and interpretable due to these limitations. To address these gaps, the following innovative steps were taken:

1.A robust preprocessing pipeline was implemented, incorporating iterative imputation using Bayesian Ridge to effectively handle missing values.

2.The approach combines robust scaling, standard scaling, and min-max scaling sequentially to scale the data while preserving its distributional integrity.

3.To enhance model accuracy and interpretability, Boruta feature selection with RF was used to identify important features.

4.A voting classifier that combines XGBoost, LightGBM, and CatBoost was suggested as a way to improve the XAI-Churn TriBoost model's prediction accuracy and reliability.

5.Using 60% for training and 40% for testing, the model was trained and evaluated on a massive dataset with over 2 million records, proving its effectiveness and robustness over a big dataset.

6.In order to interpret the model's forecasts and offer practical understanding of the variables impacting churn, explainability methods like LIME and SHAP were employed.

3. Methodology

As depicted in Fig. 1, our suggested XAI-Churn TriBoost model for forecasting customer churn adheres to a systematic process. The churn dataset is first collected. The data is then cleaned, encoded, and scaled as part of the preprocessing steps that use Bayesian ridge for iterative imputation. After that, we employ feature engineering to generate new features and select important using the Boruta technique with RF. The data is split into training and testing sets, and we balance the training set using SMOTE. We train our model using a soft voting ensemble of three boosting algorithms: XGBoost, CatBoost, and LightGBM. For interpretability, we apply explainable AI techniques, including LIME and SHAP, to understand the predictions and provide actionable insights.

3.1. Dataset

In this study, we utilized a dataset provided through the Expresso Churn Prediction Challenge on Zindi [59]. The train.csv dataset is used in this study and is accessible to

participants who join the competition on the Zindi platform. The dataset includes both numerical and categorical features, consisting of 17 variables and 2,154,048 samples. These variables capture various details about customer actions, purchases, and interactions within the network. The target variable, “churn,” indicates whether a customer becomes inactive and makes no transactions for 90 consecutive days. We use 60% of the data to train the model and 40% to test our proposed model.

3.2. Data preprocessing

Data cleaning, encoding, missing value imputation, and scaling are all part of the dataset preprocessing described in this section.

3.2.1. Data cleaning

Data cleaning is necessary since the dataset includes a large number of missing values (as illustrated in Fig. 2) and features that are not useful. We removed the following features: zone1 and zone2, which represent calls to specific zones within the telecom network, as they had over 90% missing values, making them unreliable for analysis. The moving to rivaval group (MRG) feature, indicating a client who is leaving, was also dropped because it had only one unique value. The user_id, an anonymized identifier, was removed since it does not contribute to predictive analysis. The region feature, which identifies the client's location, was excluded to make the model applicable across different countries and to avoid introducing location-based bias. top_pack, indicating the most popular package used by the customer, and freq_top_pack, recording the number of activations for the top package, were removed due to high cardinality—containing 140 categorical values—and because package types vary widely among telecom companies, limiting their general applicability. After dropping these features, we retained 11 features that are common across telecom companies for further analysis.

3.2.2. Data encoding

Data encoding is a critical step in ML, as models generally work with numerical data. Categorical features, which consist of non-numeric values, need to be transformed into a format the model can interpret. For this, we use an ordinal encoder to handle our categorical features. The ordinal encoder converts categorical values into integer labels. The model can easily process each unique category in a feature because each one is assigned a particular number. When the categories are inherently ranked, this strategy really shines. In order for the model to properly learn from categorical data, the ordinal encoder converts the values into numeric form.

3.2.3. Data imputation

Because missing values can significantly affect the quality of analysis, proper approaches are required for handling missing data. When a significant amount of data is missing, it might be quite impractical to just delete the records that have that missing information. In some instances, erasing data could result in the substantial loss of important information [20]. With little missing data, traditional methods like filling in the gaps with the mean, median, or mode are effective [18]. Nevertheless, we encounter significant missing data in this instance, since certain characteristics have missing values above 50%. In order to tackle this, iterative imputation is employed.

In statistics, iterative imputation is a technique that estimates missing values by repeatedly modeling each feature with missing data as a function of other features. By repeatedly improving the imputed values and taking advantage of interdependence, this method improves precision [39]. We mainly use a Bayesian ridge model as an estimator throughout our repeated imputation process. The coefficients of this linear regression model include prior distributions, which allows for regularization. More consistent estimations are the result, and overfitting is better controlled. By estimating uncertainty features, imputation is robust against data noise and unpredictability and is a piece of cake to implement.

Since our dataset contains outliers, we start the iterative imputation process by utilizing the median value to fill in the gaps. This aids in reducing the impact of outliers since the median is less impacted by them than the mean. Iterative imputation continues for up to twenty iterations after this first step. We guided convergence by setting an absolute tolerance of 0.00001. The method takes this tolerance level and the data scale into account to determine a scaled tolerance. As a convergence threshold, the algorithm in our situation utilized a scaled tolerance of around 18.24. After 15 iterations of iterative imputation, the imputed values stabilized within the scaled tolerance level, signifying convergence. The imputation of missing values was thorough and dependable as a consequence. In Algorithm 1, the algorithm for handling missing data is presented.

3.2.4. Data scaling

Data scaling is an essential step in ML, as it improves model accuracy and learning efficiency. Our dataset contains many outliers, so we need a careful scaling approach to handle these effectively without distorting the data distribution.

To achieve this, we apply a sequential data scaling approach [6]. First, we use robust scaling to reduce the influence of outliers. Instead of deleting these outliers, we scale them based on the median and interquartile range (IQR). This makes the scaling

less sensitive to extreme values. The formula for robust scaling is:

$$Robust\ Scaling = \frac{X - median(X)}{IQR(X)} \tag{1}$$

In this method, values are centered around the median, which minimizes the impact of outliers compared to mean-based methods.

After robust scaling, we apply standard scaling to standardize features. This scaling adjusts the data to have a mean of 0 and a standard deviation of 1. Standard scaling brings all features to a common scale, improving model performance. The formula for standard scaling is:

$$Standard\ Scaling = \frac{X - \mu}{\sigma} \tag{2}$$

Here, μ represents the mean of the feature, and σ is its standard deviation. This approach is useful for models that are sensitive to data magnitude, as it aligns each feature around a normalized range.

Finally, we apply min-max scaling to normalize the features within a fixed range, typically between 0 and 1. This ensures that all features contribute equally during model training, especially in algorithms where feature ranges are significant. The formula for min-max scaling is:

$$MinMax\ Scaling = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{3}$$

where X_{min} and X_{max} are the minimum and maximum values of the feature X, respectively. This normalization process retains the relationships within the data while compressing it to a standard range.

We successfully handle outliers and get the dataset ready for the best model performance by using our sequential scaling strategy.

3.3. Feature engineering

ML relies on feature engineering. Incorporating new features or modifying old ones is vital for improving the model. Effective feature engineering can reveal hidden relationships within the dataset. This improves the model's predictive power and makes it easier to understand. To make the model better at pattern-learning, we take useful features out of the input data and employ them.

Our database contains two distinct features: tigo and orange. The orange feature displays calls made to the orange network, whereas the tigo feature displays calls made to the tigo network. Both characteristics record the same data, which is reflective of outbound calls to different networks. We combined them into a single feature representing the overall amount of outgoing calls to external networks in order to simplify the dataset. We define this new feature as:

$$off_{net} = tigo + orange \quad (4)$$

By combining call details with this new off_net feature, we might potentially streamline the dataset. Because of this, the model is much better able to spot significant patterns. By incorporating all off-network interactions into the model, instead of just one network at a time, we make it more comprehensive.

3.4. Feature selection

Feature selection is a crucial part of ML that helps improve model performance and interpretability by extracting useful properties from the dataset. Overfitting can be prevented by enhancing model accuracy, streamlining the model, and decreasing dimensionality in tandem. The model is made easier to understand and interpret through feature selection, which simplifies it by concentrating on relevant variables. The reference is [16].

Feature selection can be done using a variety of methods; however, these methods frequently rely on thresholds or domain knowledge for manual selection, which can lead to bias. While more conventional approaches aid in feature identification, they run the risk of ignoring crucial interactions or depending on subjective judgments. The Boruta feature selection method, which compares original features with randomly produced "shadow" features, is an automated and effective strategy that we utilized to remedy issue. Boruta enhances the objectivity and reliability of the selection process by repeatedly assessing the value of traits, thereby retaining only those that are actually relevant [37]. Table 1 presents the hyperparameters and their values used in the Boruta feature selection strategy.

During each iteration, the Boruta algorithm relies on measures like the mean reduction in accuracy to determine feature relevance. It uses a tree-based classifier as its base model. We built Boruta using RF classifier because of how well its ensemble learning method assesses the relative value of features. To increase stability and decrease variance, the model was set up with 100 estimators, which means it consists of 100 DTs. The percentile of 90 ensures that only features with importance scores higher than at least 90% of the shadow features are retained. We set the significance level to 0.05, establishing a rigorous threshold for filtering out irrelevant features. To balance performance and efficiency, the maximum number of iterations was set to 100, preventing unnecessary computations. Additionally, setting the random state to 42 ensured consistent and reproducible results. This method systematically eliminates irrelevant features, retaining only the most meaningful ones. By leveraging Boruta with the RF model, we selected a reliable subset of features, enhancing the overall

performance and interpretability of our model.

Table 2 describes the selected features used for churn prediction modeling in this study. These features were selected using the Boruta feature selection method and are used as input for our model. The statistical description of these features is provided in Table 3. This statistical analysis is crucial for understanding the distribution, variability, and scale of the data, helping to identify patterns, and ensure the dataset is well-prepared for modeling.

3.5. Data splitting

Proper model evaluation and generalization depend on data splitting. The following equation represents the division of the dataset into two parts:

$$D = D_{training} + D_{testing} \tag{5}$$

In this study, we split the data with 60% allocated for training ($D_{training}$) and 40% for testing ($D_{testing}$). We chose this ratio primarily to reduce the risk of overfitting. When a model is trained on an excessively large portion of the data, it may become too specialized to the training set and fail to generalize well to new data. By allocating 40% of the data for testing, we ensure a more robust evaluation of the model's performance. Since our approach combines multiple boosting models, a larger test set helps assess generalization more effectively. Boosting models are powerful but prone to overfitting if trained on too much data without proper validation. Additionally, our dataset is highly imbalanced, with only 19% of samples representing churn cases. Using a larger test set ensures that we have enough churn samples to accurately measure model performance on minority classes.

To maintain consistency and ensure reproducibility, we set the random state to 42. We also applied stratified sampling to preserve the original class distribution in both the training and testing sets. Fig. 3 illustrates the churn distribution in the original dataset, as well as in the training and testing subsets. In Table 4, we present the statistical description of the training data, and in Table 5, we present the statistical description of the testing data. These tables provide insights into the distribution, variability, and scale of the data.

3.6. Data balancing

Imbalanced datasets are common in ML applications. When a dataset is imbalanced, the minority class represents a small fraction of the overall data. This imbalance can skew model predictions towards the majority class, thus affecting overall performance. To address this issue, several resampling techniques have been developed [30]. Traditional methods involve either duplicating samples from the minority class or

reducing the number of samples in the majority class to balance the dataset. However, these approaches can introduce problems such as overfitting or loss of valuable information, making them less effective in many scenarios. Importantly, resampling should be applied only to the training dataset. It might bias the model if applied to the whole dataset, as it would alter the real-world distribution.

The SMOTE method [13] is a popular strategy for dealing with class imbalance. By creating synthetic samples for the minority class, SMOTE avoids the overfitting danger associated with basic oversampling, which involves replicating existing samples. This method improves the model's performance on both classes by creating a training dataset that is balanced or almost balanced [11]. SMOTE creates synthetic samples by combining two close samples in the minority class, denoted as x and x_R. The nearest neighbor x_R is randomly selected from the neighbors of x within the minority class. The synthetic sample is computed as follows:

$$x_{synthetic} = x + u \times (x_R - x) \tag{6}$$

where $0 \leq u \leq 1$ is a random value.

To address the class imbalance in our dataset, we applied SMOTE using the imbalanced-learn Python library. SMOTE was implemented only on training data using a random state of 42 to ensure reproducibility. The algorithm used 5 nearest neighbors, meaning synthetic samples were generated based on the 5 closest minority class samples. The sampling strategy was set to 0.5, increasing the minority class size to 50% of the majority class. Initially, the minority class accounted for 19% of the training data, and after applying SMOTE, this proportion increased to 33%, as shown in Fig. 4. Table 6 presents the statistical summary of the training data after balancing with SMOTE. It illustrates how the distribution, variability, and scale of the data changed after the synthetic samples were generated.

3.7. Proposed XAI-Churn TriBoost

Our proposed model, XAI-Churn TriBoost, combines three boosting algorithms: XGBoost, CatBoost, and LightGBM. These algorithms are integrated into a unified soft voting ensemble to predict customer churn. Boosting algorithms are effective in handling complex patterns. Using an ensemble leverages each model's strengths to improve performance. Below is a step-by-step explanation of our methodology, shown alongside the TriBoost algorithm in Algorithm 2.

The XAI-Churn TriBoost ensemble consists of three base models. The first model, XGBoost is a popular gradient boosting algorithm known for its scalability and robust handling of overfitting through advanced regularization techniques [14]. It can handle

large datasets and non-linear relationships [44]. XGBoost uses decision trees as weak learners, optimizing them through both L1 and L2 regularization. This regularization helps prevent overfitting, making the model generalize better on unseen data. The objective function for XGBoost is defined as:

$$Obj = \sum_{i=1}^{m} l\left(y_i, \hat{y}_i^{(t-1)} + f_i(x_i)\right) + \Omega(f_k) \tag{7}$$

Here, l represents the loss function, y_i is the true label, and $\hat{y}_i^{(t-1)}$ is the prediction from the $(t-1)$-th iteration. The function output for sample x_i is denoted by $f_i(x_i)$. The regularization term, $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\omega^2$, controls complexity.

The second base model, CatBoost is specifically designed to handle categorical data efficiently without extensive preprocessing [48]. With its recurrent training technique, CatBoost enhances accuracy and stability while decreasing prediction bias through the reduction of error in each succeeding tree [24].

Furthermore, LightGBM is perfect for big datasets due to its quick training and minimal memory consumption. As opposed to traditional level-based tree-growing methods, LightGBM prioritizes the growth of the most informative leaves and grows trees leaf by leaf. Using this approach with LightGBM improves accuracy and speeds convergence while reducing the number of trees needed. In order to decrease memory usage and increase processing performance, LightGBM discretizes continuous variables into bins using a histogram-based method. With high-dimensional data in particular, this enhances performance even further [31].

We employ randomized search CV for hyperparameter tuning to guarantee that all models are operating at their optimal performance. Our extensive dataset enhances the effectiveness of our strategy. By selecting hyperparameters at random, it efficiently explores enormous parameter spaces without having to analyze every conceivable combination, as stated by [9]. We give randomized search CV a range of possible values to each model's hyperparameters and allow it to try out a certain number of combinations. This approach finds the sweet spot between exploration and efficiency, making it ideal for complicated, high-dimensional parameter fields. The final hyperparameters selected for each model are shown in Table 7.

We use a soft voting technique to integrate the predictions of each model after they have been trained. With the use of weighted probabilities, each class's model output can be determined by soft voting. In our XAI-Churn TriBoost model, the final prediction probability for class c is denoted by $P_{TriBoost}(c)$. This probability is computed by

combining the prediction probabilities from the base models using specific weights. The calculation is defined as follows:

$$P_{TriBoost}(c) = \frac{\omega_1 \cdot P_{XGBoost}(c) + \omega_2 \cdot P_{CatBoost}(c) + \omega_3 \cdot P_{LightGBM}(c)}{\omega_1 + \omega_2 + \omega_3} \quad (8)$$

In this equation, $P_{XGBoost}(c)$, $P_{CatBoost}(c)$, and $P_{LightGBM}(c)$ represent the probabilities of class c predicted by the XGBoost, CatBoost, and LightGBM models, respectively. The weights ω_1, ω_2 *and* ω_3 indicate the relative importance of each model in the ensemble. We assign weights of $\omega_1 = 2, \omega_2 = 1$ and $\omega_3 = 3$ to XGBoost, CatBoost, and LightGBM, respectively. These weights reflect the performance characteristics of each model.

The final predicted class for each sample is selected based on the highest probability from $P_{TriBoost}(c)$. This method ensures that the class with the strongest consensus across the models is chosen. This approach enhances prediction reliability and improves the overall robustness of the model.

3.8. Explainable artificial intelligence

Methods that simplify complex AI models and reveal the logic behind their predictions are part of the explainable AI area [26]. This is especially important in domains where comprehending model decisions promotes confidence and practical insights, such as customer churn prediction. We accomplish this by utilizing LIME and SHAP.

3.8.1. Local interpretable model-agnostic explanations

Complex ML models can be better understood with the help of LIME. Based on a single prediction, it builds a more straightforward and understandable model. The main concept is to make a local approximation to the model, concentrating just on the region surrounding a specific instance. This clarifies the reasoning for the prediction made by the initial model. LIME generates perturbed samples around the instance, then uses these samples to train an interpretable model such as linear model that reflects the behavior of original model near that instance [49].

LIME works by minimizing a loss function that measures the difference between the original model h and the interpretable model q around the target instance x. The approximation is weighted by proximity $\pi(x')$. The objective function is given by:

$$\delta(x) = \arg\min_{q \in Q}\big(M(h, q, \pi) + \Lambda(\mathrm{q})\big) \quad (9)$$

Here, $M(h, q, \pi)$ quantifies how well the simpler model q approximates h locally. The term $\Lambda(q)$ penalizes complexity to maintain interpretability. LIME assigns higher weights to samples closer to the target instance, ensuring locally faithful explanations.

This approach provides actionable insights across text, images, and tabular data [7].

3.8.2. Shapley additive exPlanations

SHAP is a popular explainable AI technique. It provides insights into the contribution of each feature in a model's prediction. In accordance with the principles of cooperative game theory, SHAP gives every attribute a Shapley value. The feature's contribution to the prediction is represented by this value. When it comes to sophisticated ML models, SHAP really shines since transparency is key. It treats the model like a mystery and explains both individual forecasts and the general value of features, the reference [41].

The Shapley value $\phi(j)$ for a feature j is calculated by averaging its marginal contributions across all possible feature combinations. For a set of features F and a model v, the Shapley value $\phi(j)$ is given by:

$$\phi(j) = \frac{1}{|F|!} \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \big(v(S\ U\{j\}) - v(S)\big) \tag{10}$$

In this formula, S represents each subset of F that excludes j. The term $v(S\ U\{j\}) - v(S)$ calculates the effect of adding j to subset S. A statistic for the influence of each feature is provided by SHAP by adding contributions across all combinations. This method provides interpretability on a local and global scale [3].

4. Results

4.1. Environment and setup

4.1.1. Programming language and libraries

Python was selected as the primary language for this research because to its adaptability, ease of use, and rich library support. For data analysis and manipulation, Pandas was used, while NumPy facilitated efficient numerical operations and array structures. ML models were built, data was preprocessed and imputed, and model performance was evaluated using Scikit-learn. Seaborn was used to create the data visualizations, guaranteeing that they are both aesthetically pleasing and informative. For feature selection, the Boruta library was employed to identify and retain the most relevant features. To address data imbalance, the imblearn library was utilized, specifically using SMOTE to achieve balanced class distributions.

Various boosting libraries, including LightGBM, CatBoost, and XGBoost, enabled robust model development through advanced gradient boosting techniques. A voting classifier was created using Scikit-learn to combine predictions from multiple models. For model interpretability, explainability libraries such as LIME and SHAP were utilized, providing both local and global insights into feature importance and enabling

a deeper understanding of model predictions.

4.1.2. Hardware specifications

All experiments and model training were conducted on an HP ZBook 15 G6 Mobile Workstation. The laptop is equipped with an Intel Core i9-9880H processor, an NVIDIA Quadro T2000 GPU, 32GB of DDR4 RAM, and a 512GB SSD.

4.2. Performance evaluation measures

The performance measures used in this research include accuracy, precision, recall, F1-score, and curve analysis metrics, such as area under the receiver operating characteristic curve (AUC-ROC) and the precision-recall (PR) curve. Additionally, terminologies from the confusion matrix are referenced in calculating these measures, including true positive (TP), true negative (TN), false positive (FP), and false negative (FN). For the task of churn prediction, TP represent correctly predicted churn instances, TN are correctly predicted non-churn instances, FP indicate non-churn instances incorrectly predicted as churn, and FN are churn instances incorrectly predicted as non-churn. Fig. 5 displays the confusion matrix, presenting the prediction results of the XAI-Churn TriBoost model.

4.2.1. Accuracy

Accuracy is the ratio of correctly predicted instances to the total instances. It reflects the overall prediction accuracy of the XAI-Churn TriBoost model. Accuracy is calculated using Equation:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

Fig. 6 compares the accuracy of our model with other base models. The XAI-Churn TriBoost demonstrates a higher accuracy, indicating superior performance.

4.2.2. Precision

Precision is the ratio of correctly identified churn instances to all instances predicted as churn. It measures the ability of model to avoid false positives. Precision is calculated by Equation:

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

Fig. 7 shows the precision scores of our model and the base models. The XAI-Churn TriBoost model demonstrates a higher precision, reflecting its effectiveness in minimizing false churn predictions.

4.2.3. Recall

Recall is the ratio of correctly identified churn cases to all actual churn cases. It measures the ability of model to detect all churn instances. Recall is calculated as shown

in Equation:

$$Recall = \frac{TP}{TP + FN} \tag{13}$$

Fig. 8 illustrates the recall scores of the XAI-Churn TriBoost model compared to the base models. The XAI-Churn TriBoost model shows superior recall, indicating its effectiveness in capturing most churn cases.

4.2.4. F1 score

The F1 score is the harmonic mean of precision and recall. It balances the ability of model to make precise and complete predictions. The F1 score is calculated using Equation:

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

Fig. 9 shows the F1-scores for both the base models and the XAI-Churn TriBoost model. The XAI-Churn TriBoost model achieved a high F1 score, which indicates that it effectively handles both false positives and false negatives.

4.2.5. Curve analysis

We examine the XAI-Churn TriBoost model's performance utilizing two crucial metrics from curve analysis: AUC-ROC and PR curve. The model's performance when tested against various sets of classification criteria can be better understood with the help of these indicators.

A measure of the model's performance across all thresholds is the AUC-ROC. A one-to-one change in the classification threshold has an impact on the true positive and false positive rates, as seen by the ROC curve. The model performs optimally at 1; at 0.5, it essentially makes an random estimate. Churn prediction and other imbalanced datasets benefit greatly from the PR curve, which shows the correlation between accuracy and recall at different categorization thresholds. This graph shows that the model can maintain good recall and precision even when the classes are imbalanced.

The AUC-ROC score for the XAI-Churn TriBoost model is 0.9828, and the PR AUC score is 0.9601, as illustrated in Fig. 10. The model's near-perfect results show that it can distinguish between churn and non-churn events and that the balance between recall and precision is well-maintained. This verifies that the model successfully identifies churn cases with few false positives and has good classification performance.

4.3. Explainability analysis with explainable AI techniques

Explainable AI has become crucial in understanding complex models. In applications like customer churn prediction, it is important to know why models make specific predictions. This transparency helps build trust and confidence in the model's

decisions. Traditional ML models are interpretable but often lack the predictive power needed for complex telecom data. To address this, advanced explainable AI methods like LIME and SHAP are used. These techniques explain individual predictions and highlight feature importance. They make churn prediction models more interpretable, giving stakeholders clear and actionable insights.

4.3.1. LIME analysis

Fig. 11 shows a LIME explanation for a churn prediction made by our model. On the right side, it displays the feature values for a specific instance. In the middle, it illustrates the decision-making boundary of the model. Features highlighted in blue contribute to a “no churn” prediction, while those in orange contribute to a “churn” prediction. On the left side, the model's prediction is displayed, showing a 53% probability of churn, with the actual class also being churn. Key features influencing this prediction include revenue, regularity, and arpu segment. Revenue and regularity strongly contribute towards churn, pushing the prediction in that direction. In contrast, features like arpu segment and montant slightly support a “no churn” prediction. However, the overall influence of the orange features results in the final prediction of churn.

Fig. 12 presents a LIME explanation where the model predicts no churn with a 78% probability, which matches the actual class of no churn. In this case, revenue contributes towards churn, but the combined influence of features like regularity, arpu segment, and data_volume favors no churn. These blue contributions strengthen the prediction of no churn.

4.4. SHAP analysis

Fig. 13 shows a force plot that provides an initial overview of how each feature contributes to the churn prediction for this instance. The model predicts a churn probability of 0.53. In this plot, features that increase the likelihood of churn are shown in red, while features reducing the likelihood of churn are shown in blue. For this instance, key features like ‘regularity’, and ‘revenue’ (in red) push the prediction toward churn, whereas ‘apru_segment’ (in blue) reduces the churn likelihood.

To gain a more detailed breakdown of each feature's influence, the waterfall plot in Fig. 14 is used. This plot displays the exact numerical impact of each feature on the final prediction score. Here, ‘regularity’ (+0.21) and ‘revenue’ (+0.06) are the strongest contributors to the churn prediction, while ‘apru_segment’ (-0.03) reduces it.

Together, these plots offer a comprehensive view of why the model predicts churn for this specific instance, with the force plot providing an overview and the waterfall

plot giving detailed insights into feature contributions.

On a broader level, the SHAP beeswarm plot in Fig. 15 displays the distribution of SHAP values for each feature, illustrating their impact across multiple instances. Each dot represents an instance, with its position along the x-axis indicating the SHAP value. The color gradient from blue to red reflects feature values, where red represents higher values and blue represents lower values. For example, high values of 'regularity' (in red) are associated with a strong positive impact on churn likelihood, pushing predictions toward churn. This plot reveals patterns and the direction of feature influence, with features like 'regularity', and 'montant', and having the most significant impact.

The SHAP feature importance plot in Fig. 16 ranks features based on their average absolute SHAP values, showing their overall contribution to the model's predictions. 'Regularity' and 'montant' are the top features influencing churn prediction, followed by 'arpu segment' and 'revenue'. This plot provides a clear overview of the key drivers of churn, identifying the most critical factors for model interpretation.

Taken as a whole, these SHAP summary charts shed light on the features that have a consistent impact on the model's predictions and the direction of that influence, which helps to comprehend the value of features on a local and global scale.

4.5. Ablation study

Here we show the outcomes of an ablation study that used the XAI-Churn TriBoost model to assess the effects of various weight combinations for XGBoost, CatBoost, and LightGBM. A single soft voting ensemble is created by merging these three boosting algorithms in the XAI-Churn TriBoost model. Soft voting uses a weighted probability output from each class's model to determine the final prediction. To ensure optimal performance, it is essential to evaluate the contribution of each model by varying their weights. We conducted experiments by varying the weights of XGBoost, CatBoost, and LightGBM from 1 to 3 and recorded the performance metrics for each combination. Table 8 summarizes the results of this ablation study. After analyzing the performance across all weight combinations, we selected the configuration where XGBoost: 2, CatBoost: 1, LightGBM: 3 demonstrated higher performance. This combination was chosen for the final XAI-Churn TriBoost model, as it consistently delivered superior results across multiple evaluation criteria.

5. Discussion

Customer churn is a significant problem in the TCI since retaining clients is crucial for profitability. ML has the potential to identify patterns in large datasets, enabling

companies to anticipate customer churn early. This allows businesses to take strategic actions to retain customers before they leave. ML has proven to be highly effective in making businesses more profitable in such industries. However, one of the key challenges is that ML models require large amounts of data and often function as "black boxes," making it difficult for industry professionals to fully understand their predictions. This lack of interpretability creates trust issues, making explainable AI essential.

In this research, we propose a model named XAI-Churn TriBoost, which integrates XGBoost, CatBoost, and LightGBM in a soft voting ensemble. The study is conducted on a dataset containing over two million records, with 60% of the data used for training and 40% for testing. The dataset contains a significant number of missing values, which we address using iterative imputation with Bayesian Ridge regression, instead of traditional imputation techniques. Furthermore, the dataset exhibits a high number of outliers and varying feature ranges. To handle this, we apply sequential scaling, where we first use Robust Scaling to minimize the impact of outliers, followed by Standardization, and finally, Min-Max Scaling to normalize the data within the range of 0 to 1.

Initially, the dataset consisted of 17 features, but after thorough data cleaning, removal of irrelevant features, and feature engineering, we reduced the number to 10 features. To ensure that only the most relevant and important features are retained, we employ the Boruta feature selection approach, an automated and effective method that compares original features with randomly generated "shadow" features. This process eliminates one feature, leaving us with nine selected features, which are then used for modeling. For modeling, we utilize tree-based algorithms, including XGBoost, CatBoost, and LightGBM, as they are well-suited for handling large and complex datasets. To optimize model performance, we apply Randomized Search CV for hyperparameter tuning. The predictions from these three models are then combined using a soft voting ensemble, which enhances overall accuracy and robustness.

Our proposed model outperforms individual base models, achieving an accuracy of 0.9644, precision of 0.9282, recall of 0.8782, and an F1-score of 0.9025. Additionally, we compare our approach with several state-of-the-art models, including DTs, RF, LR, Gaussian NB, Bernoulli NB, Ridge Classifier, and Stochastic Gradient Descent. The same preprocessing steps were applied to all these models to ensure a fair comparison. The results, presented in Table 9, show that Gaussian NB achieves the highest recall (0.9672) due to its probabilistic classification approach and assumption of feature

independence. This characteristic makes it more likely to classify borderline cases as positive, thereby reducing false negatives and increasing recall. However, this improvement comes at the cost of lower precision and overall accuracy.

In contrast, our proposed XAI-Churn TriBoost model delivers the best overall performance, achieving the highest accuracy (0.9644), precision (0.9282), and F1-score (0.9025). While models such as DT, Ridge Classifier, and Stochastic Gradient Descent perform well in certain aspects, they do not maintain the same level of balance across all evaluation metrics. The strength of XAI-Churn TriBoost lies in its ability to effectively identify churners while minimizing false positives, making it the most reliable and practical choice for businesses focused on customer retention strategies.

Since explainability is crucial in this domain, we incorporate LIME and SHAP. These techniques provide interpretability at both the individual prediction level and the overall model level, helping stakeholders understand the reasoning behind predictions. The SHAP analysis indicates that regularity and montant are the most important features influencing customer churn. Regularity refers to the number of activities in the last three months, while montant represents the top-up amount. Based on these findings, it appears that churn risk is highly dependent on consumer behavior patterns and transaction quantities.

In this research, our focus is solely on churn predictive modeling for TCI. However, in recent years, numerous studies have explored churn prediction across various industries, including banking and financial services, e-commerce and retail, and streaming and entertainment. These studies have applied different ML techniques to identify factors influencing customer churn and improve retention strategies.

An example of this is the churn prediction model that [5] created for a Software-as-a-Service inventory management company in Thailand. After implementing various ML models, they discovered that RFs outperformed the rest with a recall of 0.916 and an F1-score of 0.926 when utilizing 10-fold CV. Similarly, [17] constructed a predictive model to discern possible employee churners at a software company based in Portugal. According to their findings, XGBoost outperformed the other methods with ROC AUC of 0.86 and recall of 0.85.

In Bengaluru, India, [23] created a model to forecast client attrition in the online food delivery industry. They found that the NB and RFs classifiers had the best AUC score of 0.952 out of all the ML models they evaluated. According to [53], a churn prediction was developed in the banking industry using XGBoost, which resulted in an accuracy of 0.839, sensitivity of 0.601, specificity of 0.903, and F1-score of 0.613.

Similarly, [55] used LightGBM to apply several ML models and achieved accuracy of 0.91, precision of 0.94, recall of 0.87, and F1-score of 0.90.

The gaming business has also investigated churn prediction. A model for predicting player attrition in a mobile video game that is free to play was created by [43]. Their results demonstrated that the RFs model attained accuracies ranging from 0.66 to 0.95, with the range being 1-7 days for the projected churn time.

Churn prediction in the TCI sector has been the subject of multiple research, each using a unique dataset and ML method. To illustrate, Stochastic Gradient Boosting was used by [47]on 7,043 samples containing 20 features, yielding 0.981 accuracy, 0.928 precision, and 0.927 recall. In a similar vein, [35] achieved 0.981 accuracy, 0.928 precision, and 0.927 recall using an RFs model on 2,000 samples with 15 characteristics.

The discussed studies highlight the diverse applications of churn prediction models across multiple industries, each employing different ML techniques and datasets. To provide a concise comparison, Table 10 summarizes these studies, outlining their respective industries, methodologies, and key findings.

In this research, our primary focus is on churn prediction within the TCI. We utilize a single, large-scale dataset containing over 2 million records to develop and evaluate our proposed model. However, we intend to expand our analysis in subsequent work by evaluating the model's effectiveness across several industries and by integrating additional datasets. This will guarantee that our method is applicable outside of the TCI industry and assist determine its relative merits and shortcomings.

Our goal is to evaluate the model using data from other locations so that we can account for regional variances in customer behavior and further enhance its generalizability. In addition, we intend to build a web app that can link to real-time data sources and incorporate monitoring capabilities. Businesses will be able to monitor important indicators in real time, evaluate client behavior, and measure customer attrition rates with this application. This will allow them to make proactive decisions.

The application's comparative analysis features will enable businesses to categorise their clients into various categories based on demographics, product usage, and behavioral trends. This tool will assist in identifying the consumer segments most prone to subscription cancellation, enabling targeted retention strategies. We plan to implement a feedback loop enabling users to provide insights on the model's predictions to ensure ongoing improvement. The model will be periodically retrained utilising this accumulated feedback to adapt to evolving customer behaviors and enhance its predictive accuracy.

The model can stay relevant by constantly adding new data and identifying trends and patterns, so it can handle situations when external influences, such economic turmoil or changes in the market, affect consumer preferences. Additionally, automating this process will boost adaptability by improving predicted accuracy and minimising dependency on human intervention. Ultimately, this approach enhances model accuracy, which benefits decision-making, retention strategy customisation, and long-term industry reliability.

6. Conclusion

The purpose of this research was to propose a model for predicting telecom customer churn called XAI-Churn TriBoost. This ensemble model leverages soft voting and combines XGBoost, CatBoost, and LightGBM. Data quality was ensured by extensive preparation methods using a dataset of over 2 million records. With an F1-score of 0.9025, an accuracy of 0.9644, precision of 0.9282, recall of 0.8782. It was trained on 1.58 million records and tested on over 0.86 million records. Integrating explainable AI approaches improved transparency. LIME and SHAP were used to explain individual predictions and highlight feature importance, improving interpretability and building stakeholder trust. SHAP analysis revealed that regularity and montant were the most influential features affecting customer churn, providing actionable insights for retention strategies. The strong performance of XAI-Churn TriBoost underscores the importance of interpretable models in addressing churn prediction challenges.

In future work, we aim to extend the study by incorporating multiple datasets from different industries to assess the model's generalizability. Additionally, we plan to develop a web-based application with real-time monitoring to track churn rates and customer behavior, facilitating proactive decision-making. A feedback loop will also be implemented to refine predictions by periodically retraining the model with new user insights, ensuring adaptability to evolving trends and sustained predictive accuracy.

二、机器学习在电信行业客户流失预测中的综合分析

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译文：

摘要

电信行业客户流失预测代表了一个关键的商业智能任务，该任务已从主观的

人工评估演进到复杂的算法方法。在本工作中，我们提出了一种利用深度神经网络的电信客户流失预测综合框架。通过系统的问题建模、严谨的数据集分析和细致的特征工程，我们开发了一个能够捕捉客户行为中复杂模式的表明客户可能流失的模型。我们在多个性能指标上进行了广泛的实证评估，证明了我们提出的神经网络架构在现有基线方法上实现了显著的改进。我们的方法不仅提高了客户流失预测的准确性，还提供了对电信服务中导致客户流失的关键因素的可解释性见解。

术语索引: 客户流失预测、电信行业、深度神经网络

电信行业由于竞争激烈和消费者期望的变化，面临客户流失的挑战。客户流失威胁到收入和市场地位，因为获取新客户比保留现有客户花费更多。这凸显了需要先进的预测模型来识别可能流失的客户。

机器学习领域的最新进展[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22]已将客户流失预测从被动分析转变为主动干预。虽然传统方法依赖基于规则的系统和简单的统计模型，但现代深度学习架构能够捕捉客户属性与流失行为之间复杂的非线性关系。它们在多种任务中表现出卓越的性能，包括图像处理[23, 24, 25, 26]、视频理解[27, 28, 29, 30, 31]以及医学图像分析[32, 27, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42]。然而，电信领域具有独特的挑战：涵盖人口统计资料、使用模式和服务交互的异构数据类型；客户行为中的时间依赖性；以及需要可解释的预测结果以支持有针对性的客户挽留策略。

在本工作中，我们提出一个全面的电信客户流失预测框架，通过三个关键贡献来解决这些挑战。首先，我们开发了一种专门设计用于处理电信数据多模态特性的神经架构，结合了静态客户属性和动态使用模式。其次，我们引入了一种新颖的特征工程流程，从原始交易数据中提取行为指标，捕捉出表明客户流失倾向的细微模式。第三，我们通过在 IBM 电信数据集上的大量实验表明，我们的方法在保持对业务部署至关重要的可解释性的同时，实现了最先进的性能。

我们的实证分析表明，所提出的模型不仅在预测准确性上超越了现有基线，还提供了关于客户流失主要驱动因素的可操作见解。这些发现使电信提供商能够实施有针对性的挽留活动，优化资源配置，并最终将客户关系管理从被动应对转变为积极主动的范式。

II 问题建模

II-A 问题定义

我们将客户流失预测任务表述为一个二分类问题。给定一个由包含人口统计属性、服务使用模式和账户信息的特征向量 $x_i \in \mathbb{R}^d$ 表示的客户 c_i，我们的目标是学习一个函数 $f: \mathbb{R}^d \to \{0,1\}$，以预测该客户在指定时间范围内是否会流失（1）或保持活跃（0）。

正式来说，我们寻求最小化期望风险：

$$\mathcal{R}(f) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\mathcal{L}(f(x), y)] \tag{1}$$

其中 $\mathcal{D}$ 表示特征和标签的联合分布，$\mathscr{L}$ 表示损失函数。

II-B 数据集描述

我们使用包含 N=7,043 个客户记录和 d=21 个特征的 IBM 电信数据集。特征空间包括：

- 人口统计属性：年龄、性别和依赖状态
- 服务订阅：互联网、电话、流媒体和安全服务
- 账户信息：合同期限、合同类型、计费方式和月费
- 目标变量：二元流失指示器

该数据集存在类别不平衡问题，约有 26.5%的正样本（流失），因此需要仔细考虑评估指标和抽样策略。

II-C 数据预处理流程

我们的预处理流程解决了三个关键方面：

缺失值填补：我们采用零填补的方法处理缺失值，这是由于缺失值的稀疏性（¡ 2%）（此处的异常符号原文如此）以及缺失服务的分类解释。

异常值检测：我们实施基于四分位距（IQR）的异常值检测：

$$Outlier = \{xi: xi < Q1 - 1.5 \times IQR \; or \; xi > Q3 + 1.5 \times IQR\} \tag{2}$$

特征工程：分类变量使用独热编码进行编码，而连续特征则被标准化为零均值和单位方差。

II-D 模型架构

我们提出了一种前馈神经网络，其结构如下：

- 输入层: $x \in \mathbb{R}^{21}$
- 隐藏层: 三个全连接层，每层有 32 个单元，使用 ReLU 激活函数
- 输出层: 对两个类别进行 Softmax 运算

该模型在 Apple M1 处理器上使用 Adam 优化器及其默认超参数($\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)进行训练。

II-E 评估框架

考虑到业务背景和类别不平衡问题，我们采用了一个全面的评估框架：

主要指标：

- 精确率： $P = \frac{TP}{TP+FP}$ - 在误报会导致挽留活动成本时至关重要
- 召回率： $R = \frac{TP}{TP+FN}$ - 在漏掉实际流失客户会导致收入损失时必不可少
- F1 分数：$F1 = 2 \cdot \frac{P \cdot R}{P+R}$ - 在精确率和召回率之间取得平衡

业务对齐指标：我们引入了一个成本敏感的评估框架，其中误分类成本反映

了业务现实：

$$Total\ Cost = c_{FP} \cdot FP + c_{FN} \cdot FN \tag{3}$$

其中 c_{FP} 代表误判用户的保留活动成本，c_{FN} 表示未检测到流失用户所导致的收入损失。

这种表述使优化策略能够与商业目标对齐，从而促进有针对性的客户挽留活动和最优资源分配。

III 数据探索与分析

III-A 背景与上下文

本研究使用 IBM 的全面电信数据集来构建预测客户流失和挽留策略的模型。该数据集包含服务订阅信息、账户特征和人口统计数据，提供了客户与公司互动的全景视角。通过分析这些互动，我们旨在识别客户流失的模式和指标，从而支持主动的客户挽留策略。这种方法使电信服务提供商能够适应市场变化，同时关注客户需求。

III-B 研究动机

电信行业竞争激烈，保持客户对于收入和市场地位至关重要。客户流失是一个重大挑战，因为获取新客户成本是保留现有客户的五到二十五倍。本分析重点在于通过预测建模和行为分析来理解并减少客户流失。通过识别客户脱钩的关键原因，我们制定策略以提升客户忠诚度和终身价值。本研究的见解使个性化服务和针对性干预成为可能，从而在饱和的市场中增强竞争优势。

III-C 数据集特征

该数据集包含 7,043 条客户记录，每条记录有 21 个特征，详细描述了客户资料和行为。特征分为服务模式、账户属性和人口统计信息三类。服务特征涵盖电话服务、互联网类型以及在线安全、流媒体等服务。账户详情包括客户在公司的持续时间、合同类型、支付方式和账单信息。人口统计信息包括年龄、性别和家庭情况。这一丰富的特征集有助于客户细分和预测建模，识别不同客户群体中的流失指标。

III-D 特征分类与描述

数据集的结构通过以下特征类别促进了多维分析：

III-D1 目标变量

二进制流失指标表示客户是否在上个月内终止了服务，作为预测建模的主要因变量。

III-D2 服务组合

客户订阅涵盖多个服务类别。电话服务包括基础电话连接和多线路选项。互联网服务包括 DSL 和光纤连接，带宽规格各不相同。增值服务包括在线安全、云备份、设备保护和技术支持。娱乐服务包括流媒体电视和电影服务。

III-D3 账户属性

时间特征包括客户持续时间，以自首次服务激活以来的月数衡量。合同安排包括月付、年付和两年期承诺。支付方式包括电子转账、自动银行扣款、信用卡交易和传统支票支付。账单偏好表明无纸化采用率。财务指标涵盖每月重复费用和每位客户的累计收入。

III-D4 人口统计特征

客户特征包括年龄分布、性别分类、婚姻状况以及是否有依赖人口，从而实现基于人口统计学的细分和针对性的客户保留策略。

III-E 探索性数据分析

III-E1 月度费用差异分析

对平均月度费用的分析显示，流失客户与保留客户群体之间存在统计学意义上的显著差异。终止服务的客户平均月度费用为 74.44 美元，而保留客户为 61.27 美元，高出 21.5%。这种差异表明价格敏感性可能是导致客户流失的一个潜在因素，尤其是在客户对服务价值的感知无法证明更高费用的情况下。观察到的这种模式表明，可以采用基于价值的定价策略，并针对价格敏感群体提供定向折扣。图 1 展示了不同客户群体之间的费用差异。

III-E2 客户生命周期价值分布

客户生命周期价值（CLV）分析显示，流失和保留客户群体之间存在明显的差异模式。保留客户具有集中分布的 CLV，中位数较低，但包含忠诚度高、收入高的异常值。流失客户显示出更高的 CLV 中位数和更大的方差，表明有价值的客户可能由于未满足的服务期望而流失。图 2 中流失客户 CLV 的双峰分布突显了对高价值客户群体实施个性化保留策略的必要性。

III-E3 用户任期分布与留存模式

对客户 tenure 的时间分析突显了生命周期中的留存动态。初期几个月显示出高获取率和早期流失率的下降。从第 10 到第 60 个月的稳定期表明客户成功入网并感到满意。第 70 个月出现的第二个高峰标志着忠诚的长期客户。图 3 中呈现的三峰分布表明需要针对不同生命周期阶段采取定制策略：早期阶段应加强入网体验和期望管理，而中期则应关注价值和忠诚度计划。

IV 方法论

IV-A 数据预处理

有效的数据预处理是构建稳健的客户流失预测模型的基本前提。我们的预处理流程解决了三个关键挑战：分类变量编码、缺失值填补以及异常值检测。这些转换确保数据与机器学习算法的兼容性，同时保持对准确预测至关重要的基础统计特性。

预处理框架包括：

• 类别编码：将分类变量转换为模型可以处理的数值表示形式，采用保留数据内部语义关系的编码方案。

• 缺失值填补：实施基于上下文的填补策略，利用特征间的关联关系来估计缺失值，同时保持数据的完整性。

• 异常值检测与处理：应用稳健统计方法来识别并减轻可能影响模型性能的异常观测值的影响。

在这些转换之后，我们进行标准化处理，以确保特征之间的尺度不变性，防止对数值范围较大的变量产生偏差，并在模型训练过程中促进最佳收敛。

IV-A1 类别变量编码

电信数据集中类别变量的异质性需要定制化的编码策略，以保持每个特征类别内固有关系的完整性：

二进制编码：二进制分类变量包括性别、合作伙伴、依赖者、无纸化账单和电话服务，采用直接二进制映射(“Yes”→1, “No”→0)，在保持可解释性的同时实现数值计算。

服务等级的有序编码：表示服务质量等级的多类变量接受有序编码，以反映隐含的服务等级结构。对于互联网服务，我们实施映射：{“Fiber optic”→2,“DSL”→1,“No”→0}，捕捉服务能力的固有顺序。

时间归一化：合同持续时间的编码采用时间归一化以实现一致的表示：{“Two year”→2.0,“One year”→1.0,“Month-to-month”→0.083}。这种方法将合同长度标准化为年度单位，从而能够与基于任期的计算进行一致整合。

IV-A2 缺失值填补策略

我们的填补方法利用领域知识和特征间的依赖关系来系统地处理缺失数据：

基于模型的财务变量填补：TotalCharges 中的缺失值使用可观测特征的确定性关系进行填补：

$$TotalCharges_{imputed} = Contract \times Tenure \times MonthlyCharges \tag{4}$$

这种方法利用了账单变量之间的数学关系，确保填补的值与客户的账单历史保持一致。

语义零填补：对于服务相关特征，缺失值采用零填补，反映了缺失数据意味着未订阅可选服务的语义解释。这一策略符合业务背景，即未记录的服务意味着未使用。

IV-A3 异常值检测与处理

为增强模型对异常观测值的鲁棒性，我们采用四分位距（IQR）方法进行异常值识别：

$$IQR = Q_3 - Q_1 \tag{5}$$

其中 Q_1 和 Q_3 分别表示第一四分位数和第三四分位数。落在 $[Q_1 - 1.5 \times$

$IQR, Q_3 + 1.5 \times IQR]$ 范围之外的观测值被归类为异常值，并用该特征的均值进行替换，从而减轻其对模型参数的不成比例影响。

随后，我们对所有特征进行标准化处理：

$$x_{scaled} = \frac{x - \mu}{\sigma} \tag{6}$$

其中 μ 和 σ 分别表示特征的均值和标准差。这种转换确保了所有特征在模型优化过程中具有相等的贡献。

IV-B 模型架构

我们的预测框架采用了一种完全连接的神经网络架构，该架构针对客户流失预测任务进行了优化。经过广泛的架构探索，我们确定三层前馈网络对于此数据集最为理想，在模型容量与泛化性能之间取得了平衡。

网络包含三个隐藏层，每层包含 32 个神经元并使用 ReLU 激活函数，从而实现非线性特征转换，同时保持计算效率。如图 4 所示，该架构是通过系统性实验比较多种神经网络结构（包括 Transformer [43]、卷积神经网络 [44] 和 Mamba [45]）后选定的。这些更复杂的架构在我们的数据集上表现出显著的过拟合现象，这表明相对较小的数据复杂度更适合使用更简单、更容易进行正则化的模型。

IV-C 实验设置

我们使用 PyTorch 在 Apple M1 硬件上实现我们的模型，采用 Adam 优化器并使用默认超参数。训练配置包括：

- 训练-测试分割比例：80:20
- 学习率： $\alpha = 10^{-3}$
- 权重衰减正则化： $\lambda = 10^{-3}$
- 批处理大小：32
- 训练轮数：100 轮并带有早停机制

权重衰减参数作为 L2 正则化，通过惩罚大的权重幅度来防止过拟合，鼓励模型学习更通用的模式。

与基线方法的比较评估表明，我们的方法具有优越性，详见表 I。我们的模型在 Telco Customer Churn 数据集的所有评估指标上均达到了最先进的性能。

IV-D 模型评估框架

评估客户流失预测模型需要仔细考虑与业务相关的指标，以捕捉预测错误的非对称成本。我们采用一个全面的评估框架，包括：

IV-D1 准确率

虽然准确率可以提供一个总体的性能度量，但仅凭准确率不足以体现流失预测中对精确度的细致要求，因为在流失预测中，假阳性与假阴性错误具有不同的商业影响。

IV-D2 精度

高精度表明，预测将流失的 81.96%的客户实际上是会流失的客户，从而减少了可能导致不必要的挽留支出或因不当干预而使客户疏远的误报情况。

IV-D3 召回率

该模型成功识别了 82.26%的实际流失客户，这对于最大化客户保留机会和防止未被发现的客户流失导致的收入损失至关重要。

IV-D4 F1-Score

作为精确率和召回率的调和平均数，F1 分数提供了对模型性能的平衡评估，特别是在误报和误判成本相近的情况下具有特别的价值。

这些指标共同展示了模型的强大性能，其中平衡的 F1 分数表明在最大化客户流失检测的同时最小化误报的竞争目标上实现了有效的优化。

IV-E 商业影响考虑因素

评估指标的选择应与特定的商业背景和运营限制相一致：

• 精确率导向场景：当客户挽留措施成本较高或可能影响客户满意度（例如，激进的促销活动）时，优先考虑精确率可以减少资源浪费和客户困扰。

• 召回率导向场景：在高价值客户群体中，客户流失可能导致重大收入损失，此时最大化召回率可确保全面覆盖有流失风险的客户。

• 平衡方法：当干预成本和防止客户流失的收益相近时，F1-score 优化能够在竞争目标之间提供一个平衡点。

IV-F 战略商业应用

我们的模型生成的预测洞察能够支持在多个运营维度上的针对性商业策略：

IV-F1 基于风险的客户细分

通过识别高风险客户群体，企业可以实施与流失概率和客户价值相匹配的分层保留策略。这包括针对高风险群体的个性化优惠、主动的服务干预以及增强的客户服务。

IV-F2 资源优化

预测流失得分有助于高效分配挽留资源，将精力集中在高流失风险且具有较高生命周期价值的客户身上。这种优化在保持稳定客户群体服务质量的同时，最大化挽留投资的回报。

IV-F3 产品开发智能

对特征重要性和流失相关因素的分析可提供可操作的见解，以指导产品改进。了解哪些服务属性与客户留存相关，能够推动基于数据的产品开发，解决客户痛点并加强忠诚度驱动因素。

IV-F4 持续改进框架

在模型预测、干预结果和客户反馈之间实施反馈循环，能够实现预测模型和

客户保留策略的迭代优化，确保其适应客户行为和市场动态的演变。

通过系统地应用这些策略，企业可以将预测分析转化为实际的商业价值，降低客户流失率，同时提升客户满意度和运营效率。

V 结论

本文提供了项目背景、目标、范围、数据集、数据预处理步骤、模型配置和评估的初步定义。我们利用 IBM 提供的全面数据集，分析客户行为和特征以预测客户流失，并旨在开发针对性的客户保留策略。通过深入的数据探索与分析，我们识别出导致客户流失的关键因素。随后，我们采用数据预处理技术，并提出一个用于预测电信行业客户流失的模型。最后，我们评估核心指标，并根据业务需求进行解释。实验结果表明，我们提出的模型具有良好的泛化能力和准确性。

附：原文：

Abstract

Customer churn prediction in the telecommunications sector represents a critical business intelligence task that has evolved from subjective human assessment to sophisticated algorithmic approaches. In this work, we present a comprehensive framework for telecommunications churn prediction leveraging deep neural networks. Through systematic problem formulation, rigorous dataset analysis, and careful feature engineering, we develop a model that captures complex patterns in customer behavior indicative of potential churn. We conduct extensive empirical evaluations across multiple performance metrics, demonstrating that our proposed neural architecture achieves significant improvements over existing baseline methods. Our approach not only advances the state-of-the-art in churn prediction accuracy but also provides interpretable insights into the key factors driving customer attrition in telecommunications services.

Index Terms: Customer churn prediction, telecommunications industry, deep neural networks

The telecommunications industry struggles with retaining customers due to intense competition and changing consumer expectations. Customer churn threatens revenue and market standing, as acquiring new customers costs significantly more than retaining existing ones. This highlights the need for advanced predictive models to identify customers at risk of leaving.

Recent advances in machine learning [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22] have transformed churn prediction from reactive analysis to proactive intervention. While traditional approaches relied on rule-based systems and simple statistical models, modern deep learning architectures can capture complex,

non-linear relationships between customer attributes and churn behavior. They have demonstrated superior performance on diverse tasks, including image processing [23, 24, 25, 26], video understanding [27, 28, 29, 30, 31], and medical image analysis [32, 27, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42]. However, the telecommunications domain presents unique challenges: heterogeneous data types spanning demographic profiles, usage patterns, and service interactions; temporal dependencies in customer behavior; and the need for interpretable predictions that enable targeted retention strategies.

In this work, we present a comprehensive framework for telecommunications churn prediction that addresses these challenges through three key contributions. First, we develop a neural architecture specifically designed to handle the multi-modal nature of telecommunications data, incorporating both static customer attributes and dynamic usage patterns. Second, we introduce a novel feature engineering pipeline that extracts behavioral indicators from raw transactional data, capturing subtle patterns indicative of churn propensity. Third, we demonstrate through extensive experiments on the IBM telecommunications dataset that our approach achieves state-of-the-art performance while maintaining interpretability crucial for business deployment.

Our empirical analysis reveals that the proposed model not only surpasses existing baselines in predictive accuracy but also provides actionable insights into the primary drivers of customer attrition. These findings enable telecommunications providers to implement targeted retention campaigns, optimize resource allocation, and ultimately transform customer relationship management from a reactive to a proactive paradigm.

II Problem Formulation

II-A Problem Definition

We formulate the customer churn prediction task as a binary classification problem. Given a customer c_i represented by a feature vector $x_i \in \mathbb{R}^d$ encompassing demographic attributes, service usage patterns, and account information, our objective is to learn a function $f: \mathbb{R}^d \to \{0,1\}$ that predicts whether the customer will churn (1) or remain active (0) within a specified time horizon.

Formally, we seek to minimize the expected risk:

$$\mathcal{R}(f) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\mathcal{L}(f(x), y)] \tag{1}$$

where $\mathcal{D}$ represents the joint distribution of features and labels, and $\mathscr{L}$ denotes the loss function.

II-B Dataset Description

We utilize the IBM telecommunications dataset comprising N=7,043 customer records with d=21 features. The feature space includes:

- Demographic attributes: Age, gender, and dependent status

- Service subscriptions: Internet, phone, streaming, and security services
- Account information: Tenure, contract type, billing method, and monthly charges
- Target variable: Binary churn indicator

The dataset exhibits class imbalance with approximately 26.5% positive (churn) instances, necessitating careful consideration of evaluation metrics and sampling strategies.

II-C Data Preprocessing Pipeline

Our preprocessing pipeline addresses three critical aspects:

Missing Value Imputation: We employ zero-imputation for missing values, justified by the sparse nature of missingness (¡ 2%) and the categorical interpretation of absent services.

Outlier Detection: We implement interquartile range (IQR) based outlier detection:

$$Outlier = \{xi: xi < Q1 - 1.5 \times IQR \; or \; xi > Q3 + 1.5 \times IQR\} \quad (2)$$

Feature Engineering: Categorical variables are encoded using one-hot encoding, while continuous features are standardized to zero mean and unit variance.

II-D Model Architecture

We propose a feedforward neural network with the following architecture:

- Input Layer: $x \in \mathbb{R}^{21}$
- Hidden Layers: Three fully connected layers with 32 units each, ReLU activation
- Output Layer: Softmax over 2 classes

The model is trained using the Adam optimizer with default hyperparameters $(\alpha = 0.001, \beta_1 = 0.9, \beta_2 = 0.999)$ on an Apple M1 processor.

II-E Evaluation Framework

Given the business context and class imbalance, we employ a comprehensive evaluation framework:

Primary Metrics:

- Precision: $P = \frac{TP}{TP+FP}$ - Critical when false alarms incur retention campaign costs
- Recall: $R = \frac{TP}{TP+FN}$ - Essential when missing actual churners results in revenue loss
- F1-Score: $F1 = 2 \cdot \frac{P \cdot R}{P+R}$ - Balances precision-recall trade-off

Business-Aligned Metrics: We introduce a cost-sensitive evaluation framework where misclassification costs reflect business realities:

where c_{FP} represents retention campaign costs for false positives and c_{FN} denotes lost revenue from undetected churners.

This formulation enables optimization strategies aligned with business objectives, facilitating targeted retention campaigns and optimal resource allocation.

III Data Exploration and Analysis

III-A Background and Context

This study uses IBM's comprehensive telecommunications dataset to create predictive models for customer churn and retention strategies. The dataset includes service subscriptions, account characteristics, and demographics, offering a holistic view of customer-company interactions. By analyzing these interactions, we aim to identify patterns and indicators of customer attrition, aiding proactive retention strategies. This approach allows telecommunications providers to adapt to market changes while focusing on customer needs.

III-B Research Motivation

The telecommunications industry is highly competitive, where retaining customers is crucial for revenue and market position. Customer churn is a significant challenge since acquiring new customers costs five to twenty-five times more than retaining existing ones. This analysis focuses on understanding and reducing customer attrition using predictive modeling and behavioral analysis. By identifying key reasons for customer disengagement, we develop strategies to enhance loyalty and lifetime value. Insights from this research enable personalized services and targeted interventions, strengthening competitive advantage in a saturated market.

III-C Dataset Characteristics

The dataset includes 7,043 customer records with 21 features each, detailing customer profiles and behaviors. Features are categorized into service patterns, account attributes, and demographics. Service features cover telephony, internet types, and services like online security and streaming. Account details include tenure, contract, payment, and billing. Demographics include age, gender, and household. This extensive feature set aids in segmentation and predictive modeling, identifying churn indicators across diverse customer groups.

III-D Feature Categories and Descriptions

The dataset structure facilitates multi-dimensional analysis through the following feature categories:

III-D1 Target Variable

The binary churn indicator denotes whether a customer terminated services within the preceding month, serving as the primary dependent variable for predictive modeling.

III-D2 Service

Portfolio Customer subscriptions span multiple service categories. Telephony services include basic phone connectivity and multi-line options. Internet services comprise DSL and fiber optic connections with varying bandwidth specifications. Value-added services encompass online security, cloud backup, device protection, and technical support. Entertainment offerings include streaming television and movie services.

III-D3 Account Attributes

Temporal features include customer tenure, measured in months since initial service activation. Contractual arrangements vary between month-to-month, annual, and biennial commitments. Payment mechanisms include electronic transfers, automated bank withdrawals, credit card transactions, and traditional check payments. Billing preferences indicate paperless adoption rates. Financial metrics capture monthly recurring charges and cumulative revenue per customer.

III-D4 Demographic Profile

Customer characteristics include age distribution, gender classification, partnership status, and dependent presence, enabling demographic-based segmentation and targeted retention strategies.

III-E Exploratory Data Analysis

III-E1 Monthly Charge Differential Analysis

Analysis of average monthly charges reveals a statistically significant difference between churned and retained customer segments. Customers who terminated services exhibited mean monthly charges of $74.44, compared to $61.27 for retained customers, representing a 21.5% premium. This disparity suggests price sensitivity as a potential churn driver, particularly when service value perception fails to justify higher costs. The observed pattern indicates opportunities for value-based pricing strategies and targeted discounts for price-sensitive segments. Figure 1 illustrates this charge differential across customer segments.

III-E2 Customer Lifetime Value Distribution

Customer lifetime value (CLV) analysis shows distinct patterns between churned and retained segments. Retained customers have a concentrated CLV distribution with lower medians but include loyal, high-revenue outliers. Churned customers show higher median CLV and greater variance, indicating that valuable customers are likely to leave due to unmet service expectations. The bimodal distribution of churned customer CLV in Fig. 2 highlights the need for personalized retention strategies for

high-value segments.

III-E3 Tenure Distribution and Retention Patterns

Temporal analysis of customer tenure highlights retention dynamics in the lifecycle. Initial months show high acquisition and early-stage churn decay. Stabilization from months 10-60 denotes successful onboarding and satisfaction. A secondary peak at month 70 marks loyal, long-term customers. This trimodal distribution presented in Fig. 3 suggests distinct lifecycle phases requiring tailored strategies: early-stage retention should enhance onboarding and expectations, while mid-tenure strategies focus on value and loyalty programs.

IV Methodology

IV-A Data Preprocessing

Effective data preprocessing constitutes a fundamental prerequisite for developing robust predictive models in customer churn analysis. Our preprocessing pipeline addresses three critical challenges: categorical variable encoding, missing value imputation, and outlier detection. These transformations ensure data compatibility with machine learning algorithms while preserving the underlying statistical properties essential for accurate prediction.

The preprocessing framework encompasses:

- Categorical Encoding: Transformation of categorical variables into numerical representations suitable for model ingestion, employing encoding schemes that preserve semantic relationships within the data.
- Missing Value Imputation: Implementation of context-aware imputation strategies that leverage inter-feature relationships to estimate missing values while maintaining data integrity.
- Outlier Detection and Treatment: Application of robust statistical methods to identify and mitigate the influence of anomalous observations that could compromise model performance.

Following these transformations, we apply standardization to ensure scale invariance across features, preventing bias toward variables with larger numerical ranges and facilitating optimal convergence during model training.

IV-A1 Categorical Variable Encoding

The heterogeneous nature of categorical variables in the telecommunications dataset necessitates tailored encoding strategies that preserve the inherent relationships within each feature category:

Binary Encoding: Binary categorical variables including Gender, Partner,

Dependents, PaperlessBilling, and PhoneService undergo direct binary mapping ("Yes"→1, "No"→0), maintaining interpretability while enabling numerical computation.

Ordinal Encoding for Service Tiers: Multi-class variables representing service quality tiers receive ordinal encodings that reflect implicit service hierarchies. For InternetService, we implement the mapping:
{"Fiber optic"→2,"DSL"→1,"No"→0}, capturing the inherent ordering of service capabilities.

Temporal Normalization: Contract duration encoding employs temporal normalization to facilitate consistent representation: {"Two year"→2.0,"One year"→1.0,"Month-to-month"→0.083}. This approach standardizes contract lengths to annual units, enabling coherent integration with tenure-based calculations.

IV-A2 Missing Value Imputation Strategy

Our imputation methodology leverages domain knowledge and inter-feature dependencies to address missing data systematically:

Model-Based Imputation for Financial Variables: Missing values in TotalCharges are imputed using a deterministic relationship with observable features:

$$TotalCharges_{imputed} = Contract \times Tenure \times MonthlyCharges \tag{4}$$

This approach exploits the mathematical relationship between billing variables, ensuring imputed values maintain consistency with the customer's billing history.

Semantic Zero Imputation: For service-related features, missing values receive zero imputation, reflecting the semantic interpretation that absence of data indicates non-subscription to optional services. This strategy aligns with the business context where unrecorded services imply non-utilization.

IV-A3 Outlier Detection and Treatment

To enhance model robustness against anomalous observations, we employ the Interquartile Range (IQR) method for outlier identification:

$$IQR = Q_3 - Q_1 \tag{5}$$

where Q_1 and Q_3 represent the first and third quartiles, respectively. Observations falling outside the bounds $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$ are classified as outliers and replaced with the feature mean, mitigating their disproportionate influence on model parameters.

Subsequently, we apply standardization across all features:

$$x_{scaled} = \frac{x - \mu}{\sigma} \tag{6}$$

where μ and σ denote the feature mean and standard deviation, respectively.

This transformation ensures equal contribution of all features during model optimization.

IV-B Model Architecture

Our predictive framework employs a fully connected neural network architecture optimized for the customer churn prediction task. Following extensive architectural exploration, we identified a three-layer feedforward network as optimal for this dataset, balancing model capacity with generalization performance.

The network comprises three hidden layers, each containing 32 neurons with ReLU activation functions, enabling non-linear feature transformations while maintaining computational efficiency. This architecture, illustrated in Fig. 4, was selected through systematic experimentation comparing various neural architectures including Transformers [43], Convolutional Neural Networks [44], and Mamba [45]. These more complex architectures exhibited significant overfitting on our dataset, suggesting that the relatively modest data complexity favors simpler, more regularizable models.

IV-C Experimental Setup

We implement our model using PyTorch on Apple M1 hardware, employing the Adam optimizer with default hyperparameters. The training configuration includes:

- Train-test split ratio: 80:20
- Learning rate: $\alpha = 10^{-3}$
- Weight decay regularization: $\lambda = 10^{-3}$
- Batch size: 32
- Training epochs: 100 with early stopping

The weight decay parameter serves as L2 regularization, preventing overfitting by penalizing large weight magnitudes and encouraging the model to learn more generalizable patterns.

Comparative evaluation against baseline methods demonstrates the superiority of our approach, as detailed in Table I. Our model achieves state-of-the-art performance across all evaluation metrics on the Telco Customer Churn dataset.

IV-F Strategic Business Applications

The predictive insights generated by our model enable targeted business strategies across multiple operational dimensions:

IV-F1 Risk-Based Customer Segmentation

By identifying high-risk customer cohorts, businesses can implement tiered retention strategies proportional to churn probability and customer value. This includes

personalized offers, proactive service interventions, and enhanced customer support for at-risk segments.

IV-F2 Resource Optimization

Predictive churn scores facilitate efficient allocation of retention resources, concentrating efforts on customers exhibiting both high churn risk and substantial lifetime value. This optimization maximizes return on retention investments while maintaining service quality for stable customer segments.

IV-F3 Product Development Intelligence

Analysis of feature importance and churn correlates provides actionable insights for product enhancement. Understanding which service attributes correlate with retention enables data-driven product development that addresses customer pain points and reinforces loyalty drivers.

IV-F4 Continuous Improvement Framework

Implementation of feedback loops between model predictions, intervention outcomes, and customer responses enables iterative refinement of both predictive models and retention strategies, ensuring adaptation to evolving customer behaviors and market dynamics.

Through systematic application of these strategies, organizations can transform predictive analytics into tangible business value, reducing churn rates while enhancing customer satisfaction and operational efficiency.

V CONCLUSION

In this paper, we provide a preliminary definition of the project's background, objectives, scope, dataset, data preprocessing steps, model configuration, and evaluation. Leveraging a comprehensive dataset provided by IBM, we analyze customer behavior and characteristics to predict customer churn, with the aim of developing targeted customer retention strategies. Through in-depth exploration and analysis of the data, we identify key factors that contribute to customer churn. Subsequently, we employ data preprocessing techniques and propose a model to predict churn in the telecommunications industry. Finally, we evaluate the core metrics and provide interpretations in alignment with business requirements. Experimental results demonstrate that our proposed model exhibits good generalization and accuracy.